

WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS2 18/11/2025

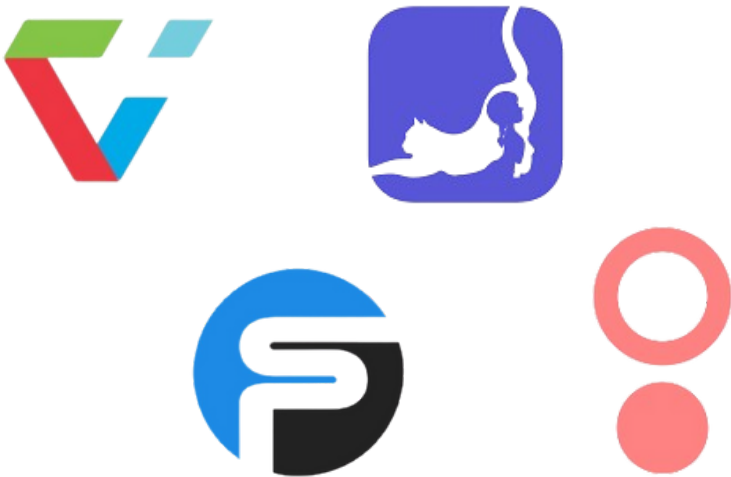


Beatriz Francisco N° 118638
Catarina Ribeiro N° 119467
Mariana Marques N°118971
Marta Cruz N° 119572
Matilde Rodrigues N°119714

MOTION SUIT

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies



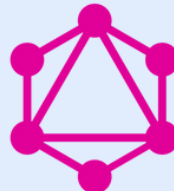


RELATED WORK

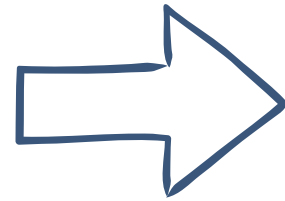
	SmartPosture	FlexiTrace	SwordHealth	DorsaVI	Current Suit	MotionSuit
Performance Analytics	✓	✓	✓	✓	✓	✓
Real-Time Feedback	✗	✗	✓	✓	✓	✓
Wearable Sensors	✗	✗	✗	✓	✓	✓
Gamification	✗	✓	✗	✗	✗	✓



RELATED WORK

	SmartPosture	FlexiTrace	SwordHealth	DorsaVI
Language	  	 	 	 
Frameworks		 	 	
Cloud & Infrastructure				
APIs & Integrations				

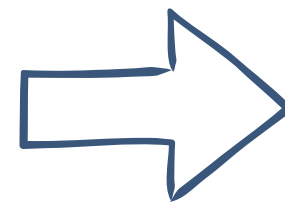
USER STORIES



As a
Suit User

I want
To receive alerts
about my posture

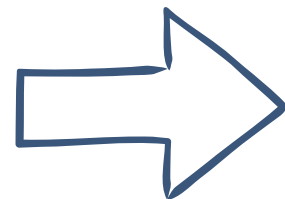
So that
I can prevent injuries in
the future



Researcher

To analyze the
captured data

I can identify postural
deviations and abnormal
patterns



Administrator

To manage user
accounts

I can control access and
provide support

FUNCTIONAL



SUIT USER

- 01** The system must allow to register and log in with their respective role
- 02** The system must deliver posture-related alerts to users
- 03** The system must provide a dashboard displaying all personal user data
- 04** The system must enable users to export their personal data

FUNCTIONAL



RESEARCHER

05 The system must display all user data with multiple views and selectable time periods.

06 The system must allow researchers to export data from all users.

FUNCTIONAL



ADMINISTRATOR

- 07** The system must provide user account management capabilities
- 08** The system must provide an administrative dashboard displaying system metrics.

NON-FUNCTIONAL

Security and Privacy

- 01** The system must encrypt all the Protected Health Information
- 02** The system must store passwords using salted cryptographic hashes

Reliability

Usability and
User Experience

Data
Management

Maintainability
and Scalability

NON-FUNCTIONAL

Reliability

Security and Privacy

- 03** The system must guarantee zero data loss at the end of each session
- 04** The system must support offline operability through local data caching

Usability and User Experience

Data Management

Maintainability and Scalability

NON-FUNCTIONAL

Usability and User Experience

Security
and Privacy

Reliability

05 The system must allow new users to perform basic monitoring tasks

06 The system must enable users to personalize dashboard layouts

Data
Management

Maintainability
and Scalability

NON-FUNCTIONAL

Data Management

Security
and Privacy

Reliability

Usability and
User Experience

07 The system must ensure high-performance queries for user-facing operations

08 The system must provide reliable retention of raw sensor data

Maintainability
and Scalability

NON-FUNCTIONAL

Security
and Privacy

Reliability

Usability and
User Experience

Data
Management

Maintainability
and Scalability

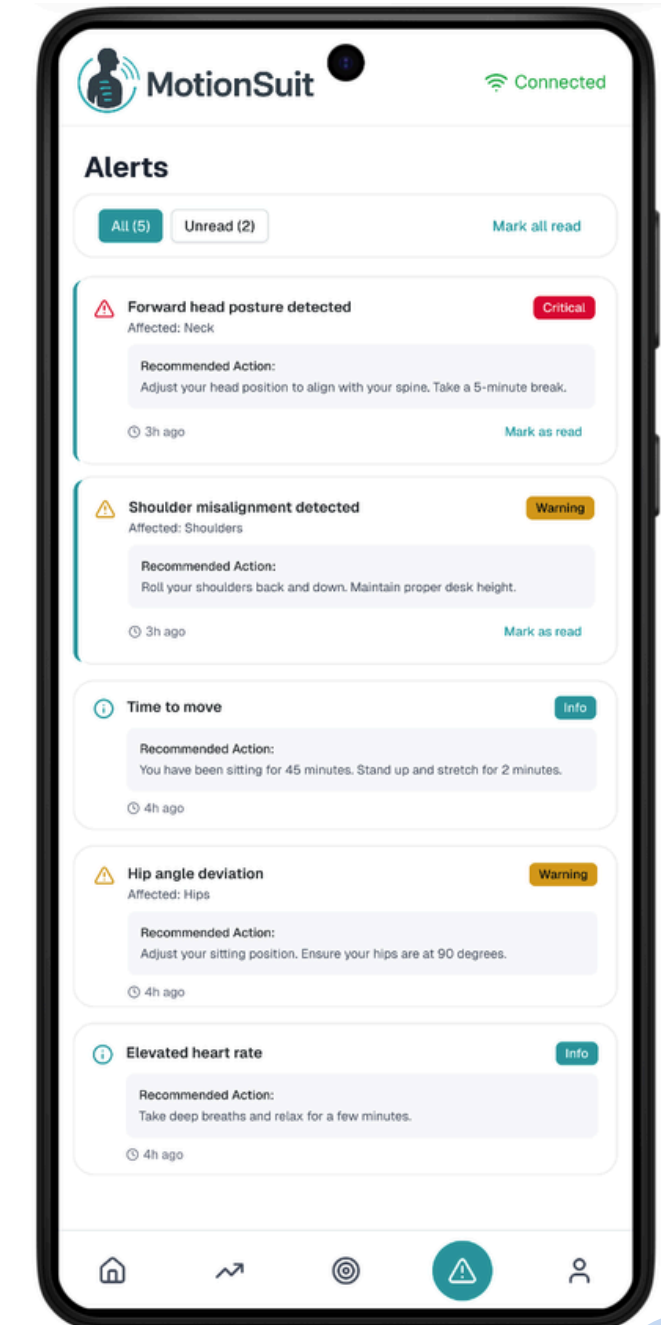
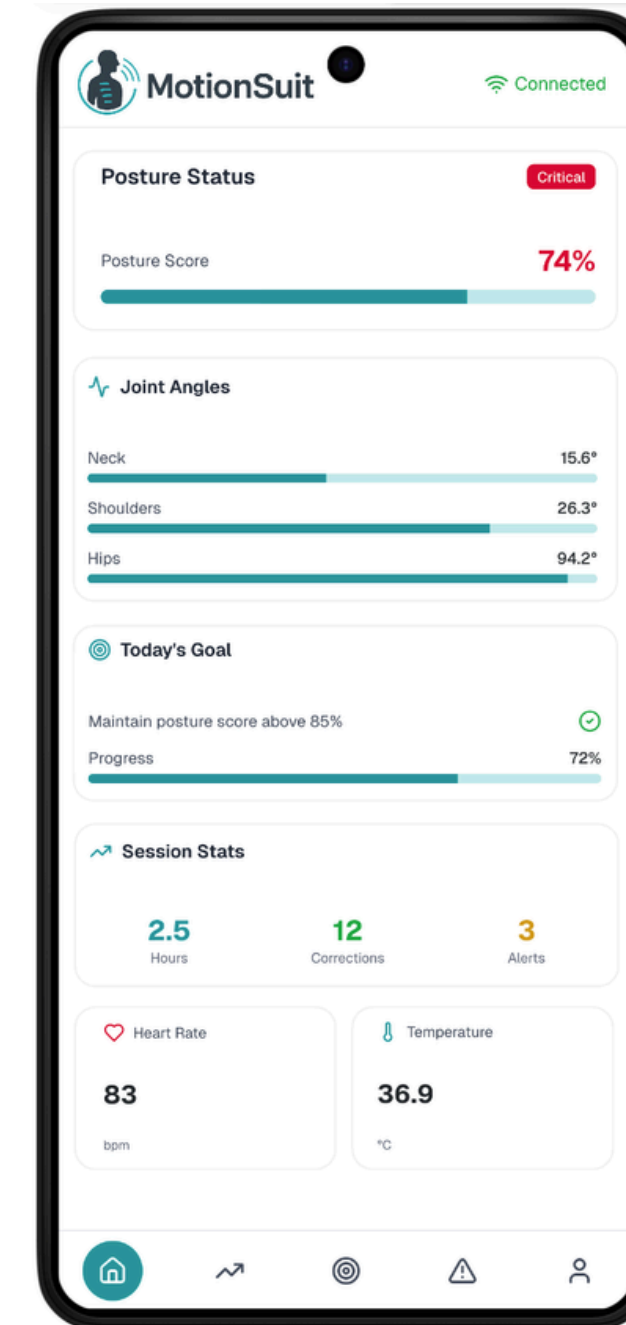
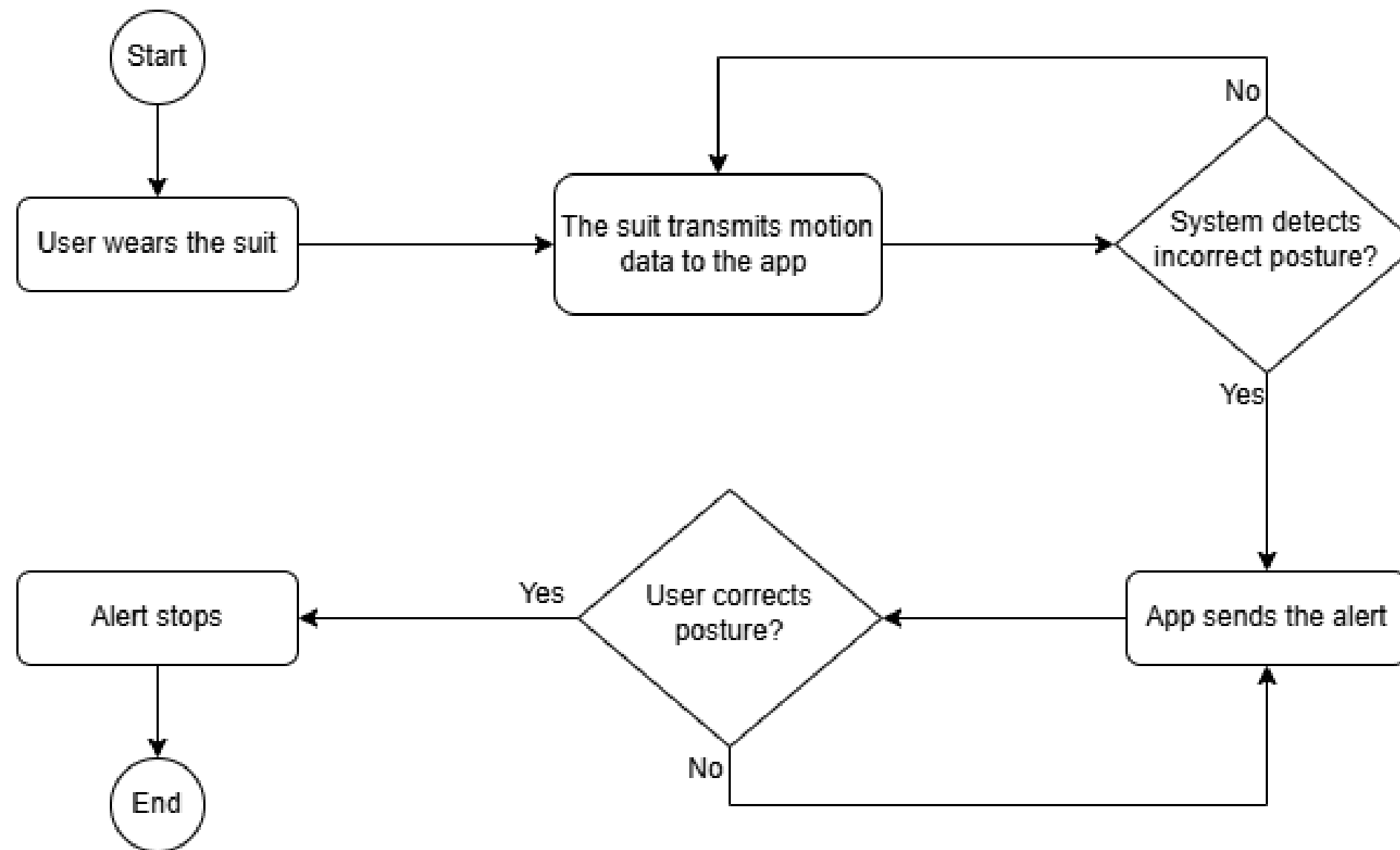
09

The system must expose well-defined APIs for all components

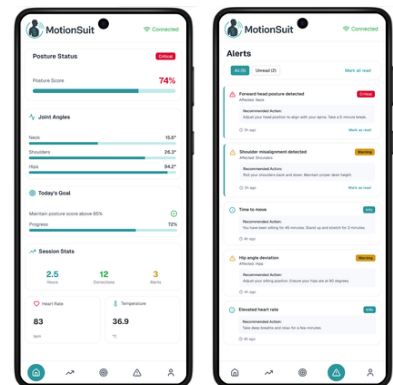
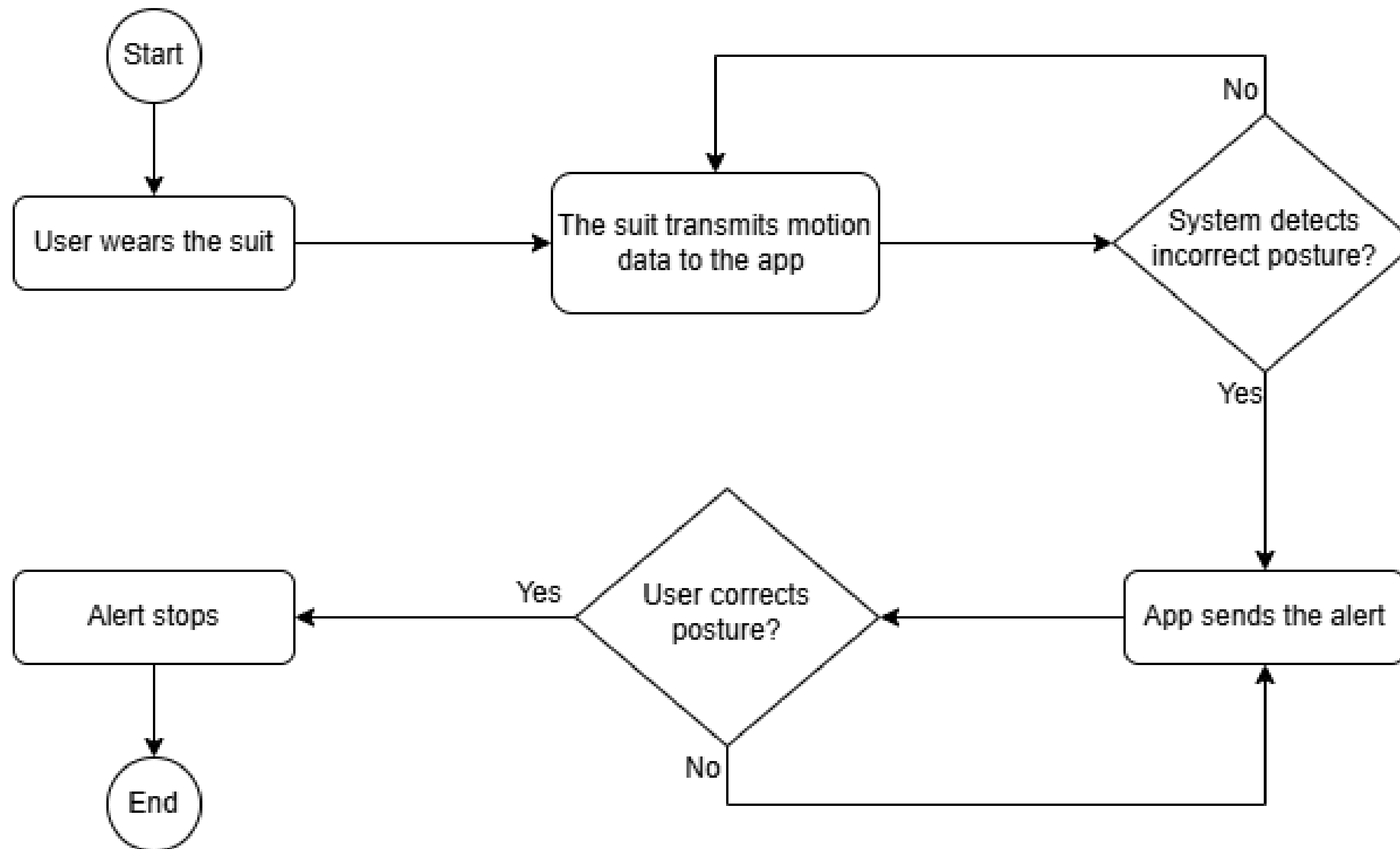
10

The system must support the addition of new sensor types through its sensor integration

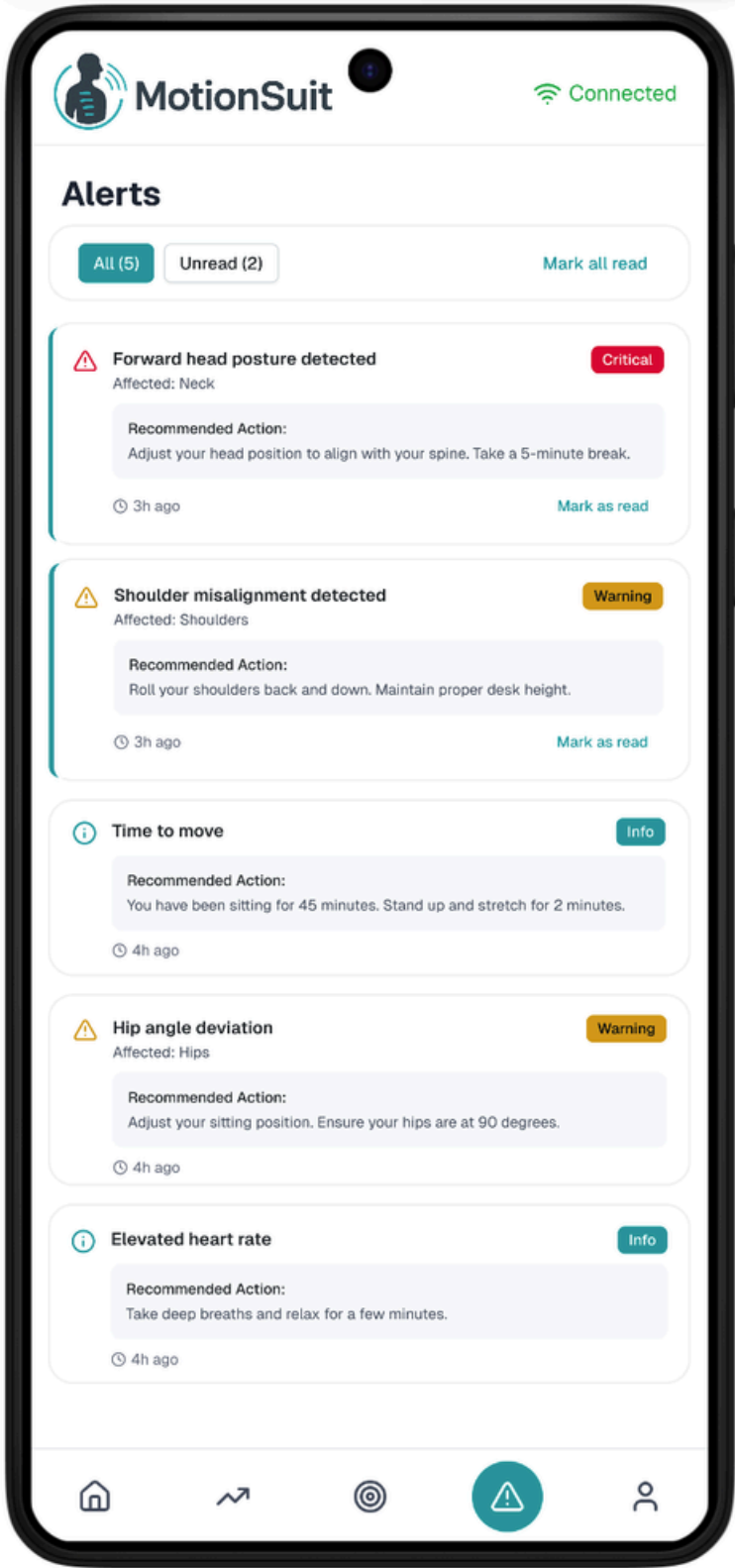
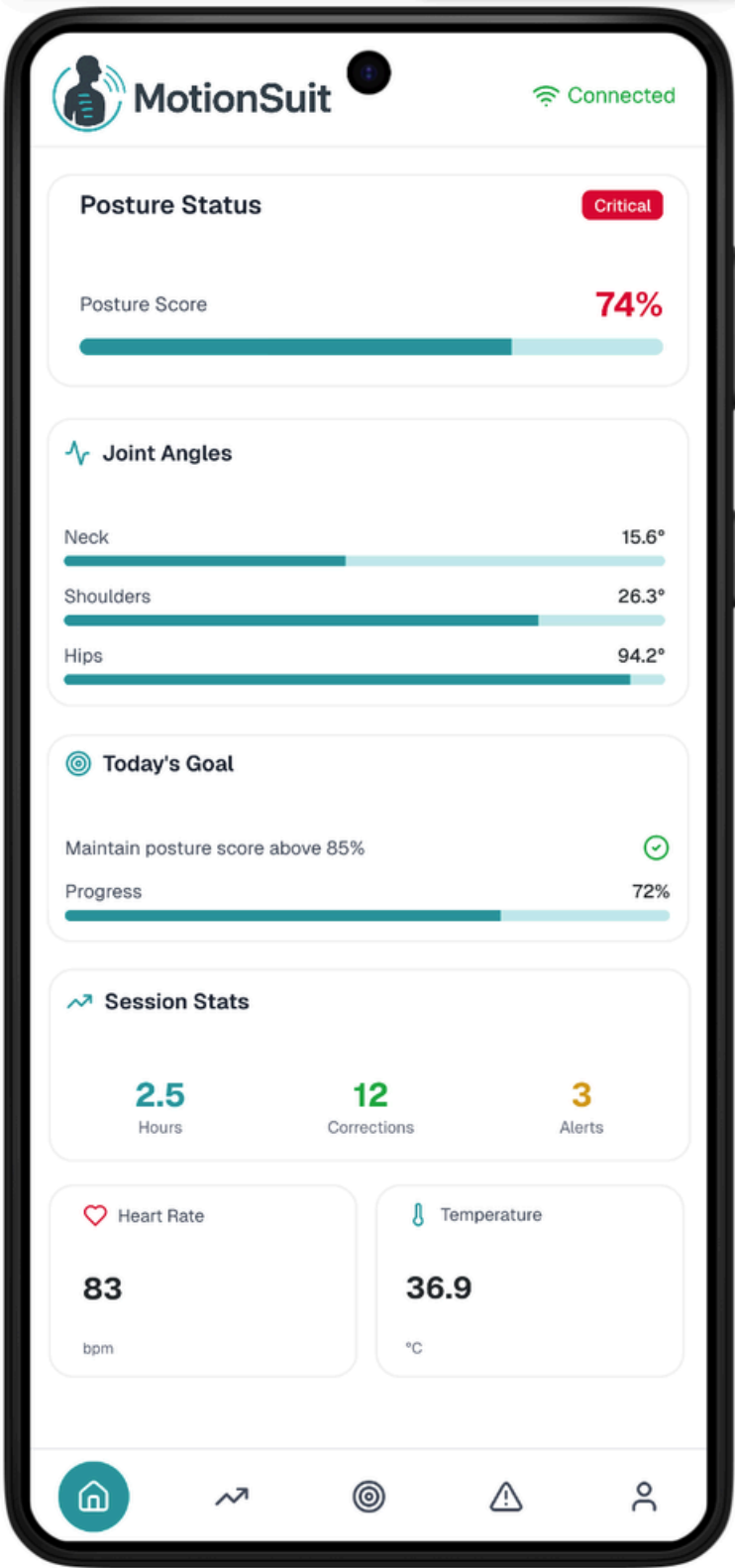
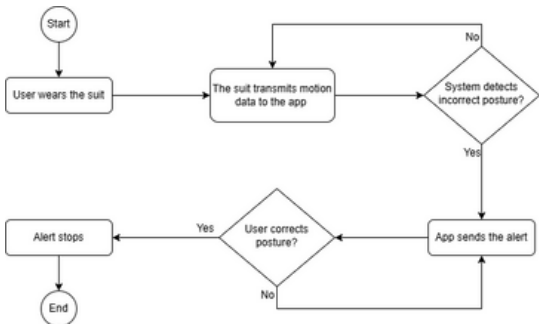
UC-1 → Receive Posture Alerts



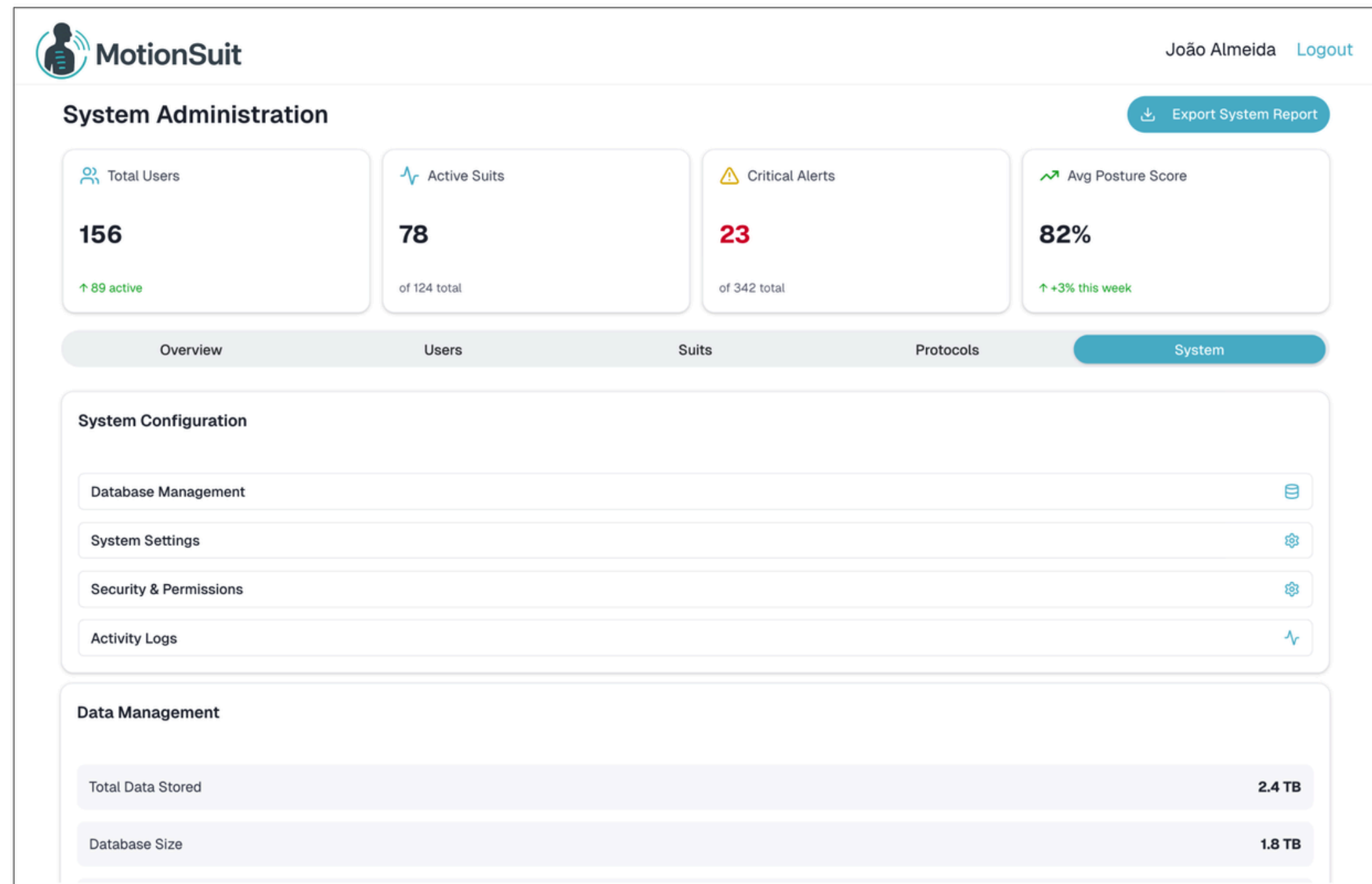
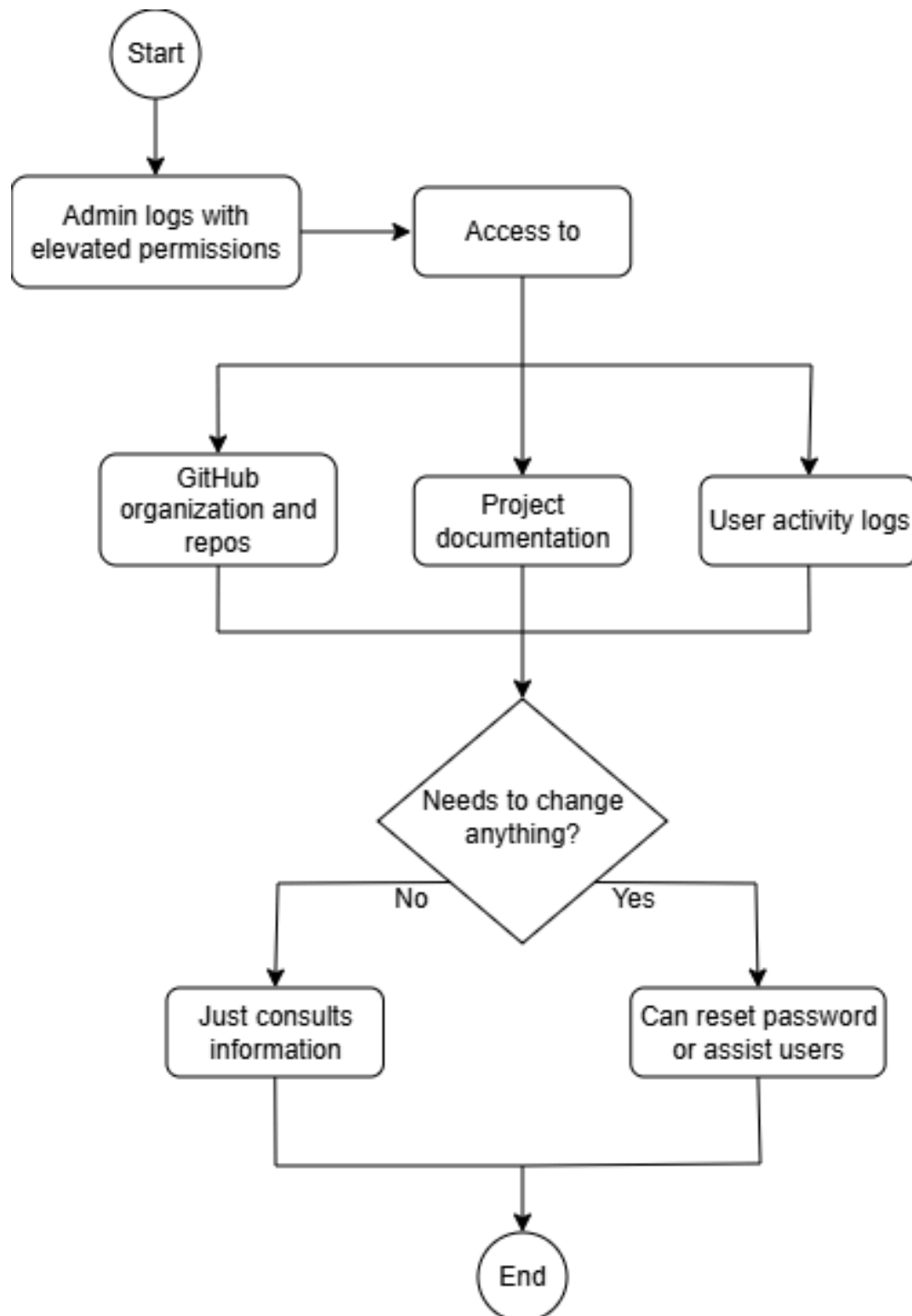
UC-1 → Receive Posture Alerts



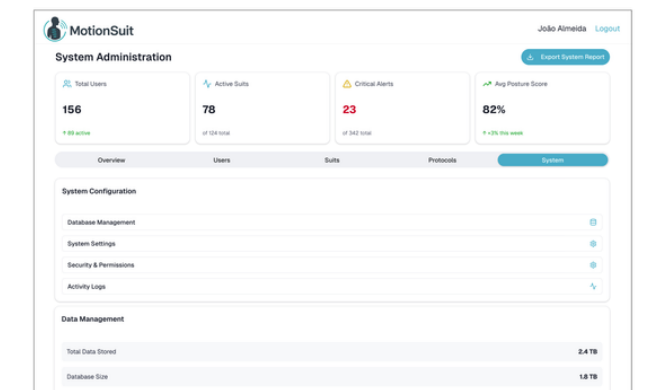
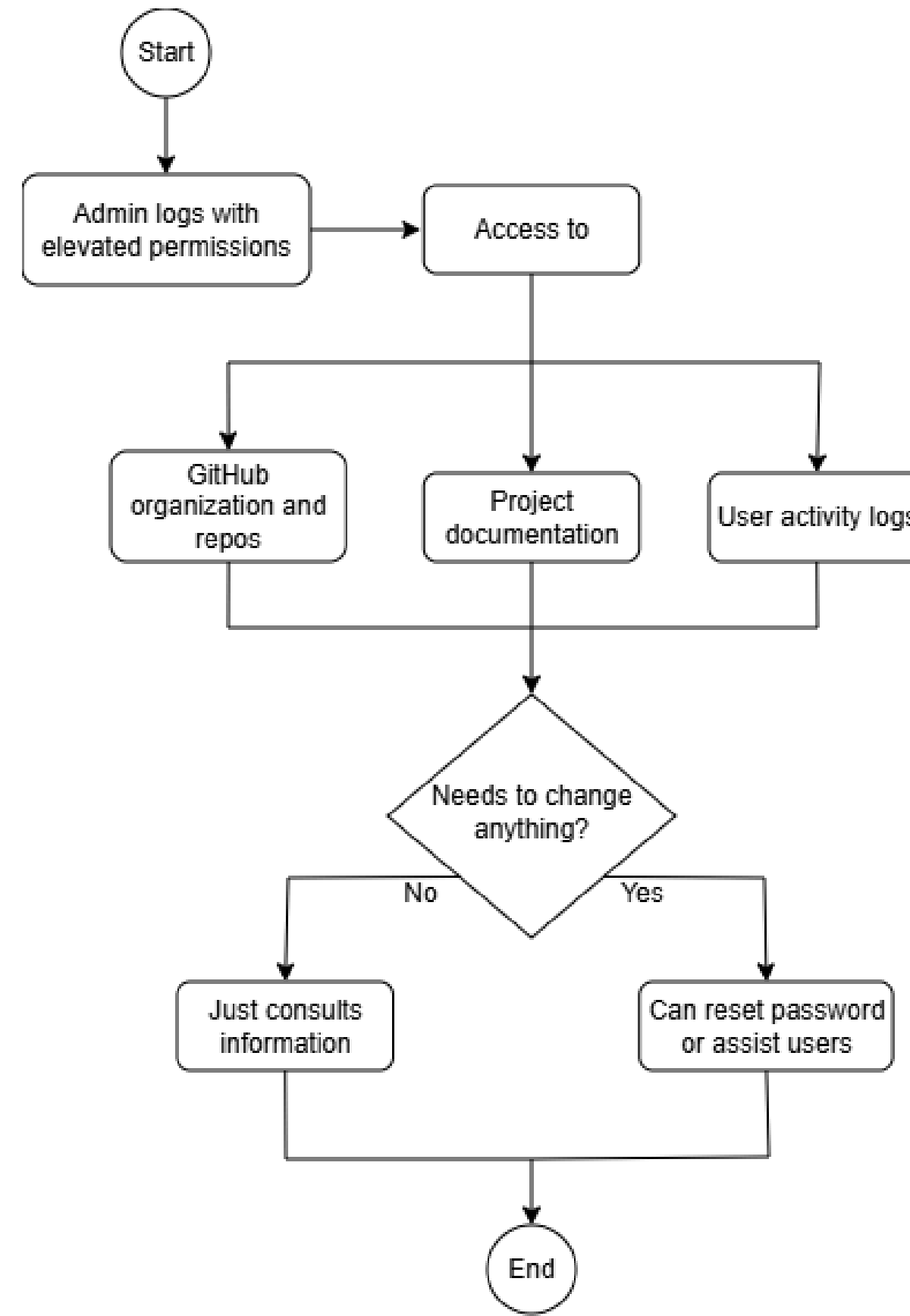
UC-1 → Receive Posture Alerts



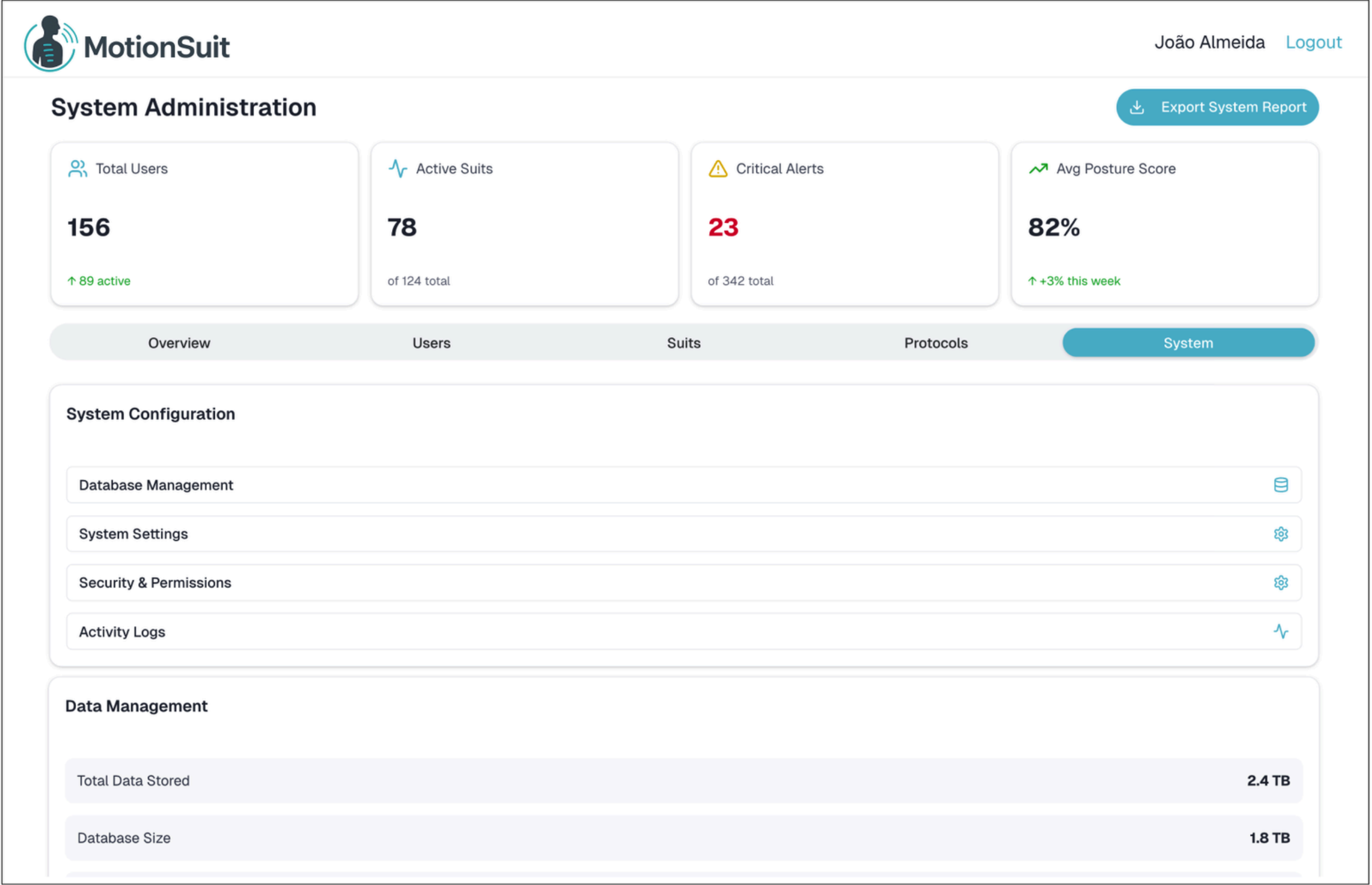
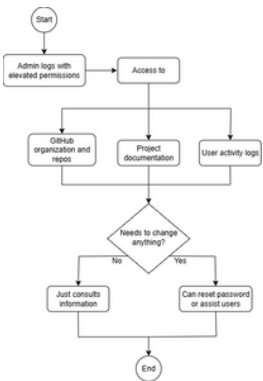
UC-9 → Access Full System Information



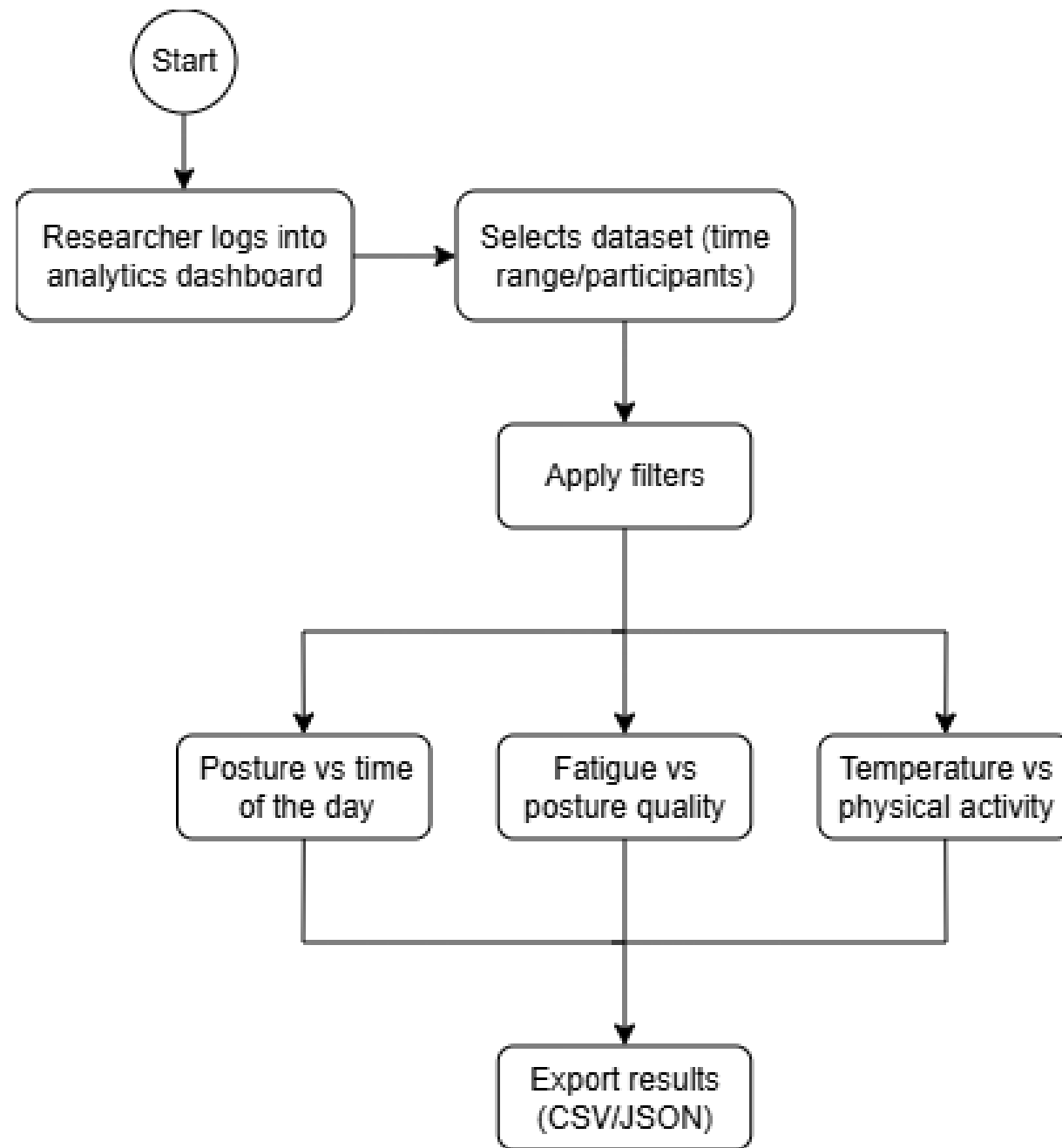
UC-9 → Access Full System Information



UC-9 → Access Full System Information

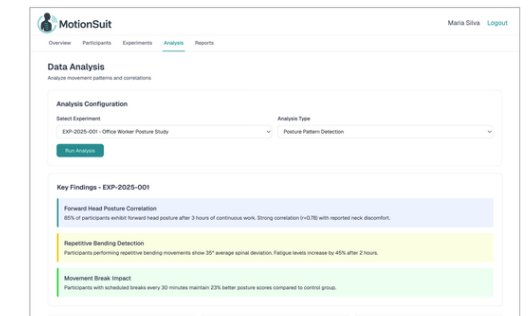
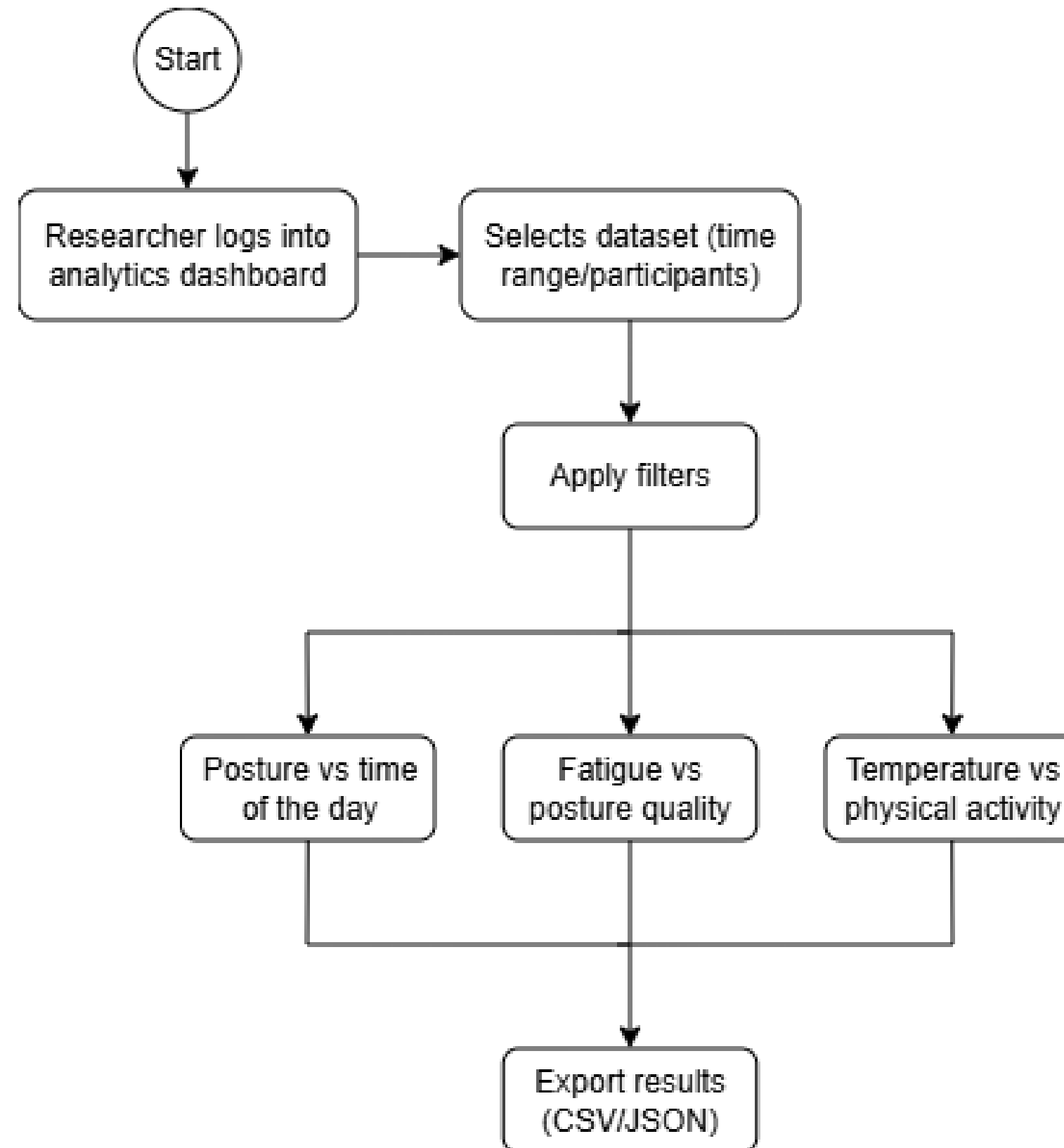


UC-11 → Analyse Captured Data

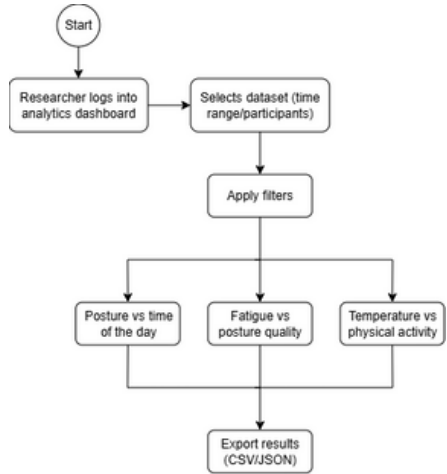


The screenshot shows the MotionSuit web application interface. The top navigation bar includes 'Overview', 'Participants', 'Experiments', 'Analysis' (active), and 'Reports'. The user 'Maria Silva' is logged in, with a 'Logout' link. The main section is titled 'Data Analysis' with the subtitle 'Analyze movement patterns and correlations'. Under 'Analysis Configuration', there is a 'Select Experiment' dropdown menu showing 'EXP-2025-001 - Office Worker Posture Study' and an 'Analysis Type' dropdown menu showing 'Posture Pattern Detection'. A 'Run Analysis' button is present. Below this, the 'Key Findings - EXP-2025-001' section displays three findings: 'Forward Head Posture Correlation' (85% of participants exhibit forward head posture after 3 hours of continuous work. Strong correlation ($r=0.78$) with reported neck discomfort.), 'Repetitive Bending Detection' (Participants performing repetitive bending movements show 35° average spinal deviation. Fatigue levels increase by 45% after 2 hours.), and 'Movement Break Impact' (Participants with scheduled breaks every 30 minutes maintain 23% better posture scores compared to control group.).

UC-11 → Analyse Captured Data



UC-11 → Analyse Captured Data



MotionSuit

Maria Silva [Logout](#)

OverviewParticipantsExperimentsAnalysisReports

Data Analysis

Analyze movement patterns and correlations

Analysis Configuration

Select Experiment

EXP-2025-001 - Office Worker Posture Study

Analysis Type

Posture Pattern Detection

Run Analysis

Key Findings - EXP-2025-001

Forward Head Posture Correlation

85% of participants exhibit forward head posture after 3 hours of continuous work. Strong correlation ($r=0.78$) with reported neck discomfort.

Repetitive Bending Detection

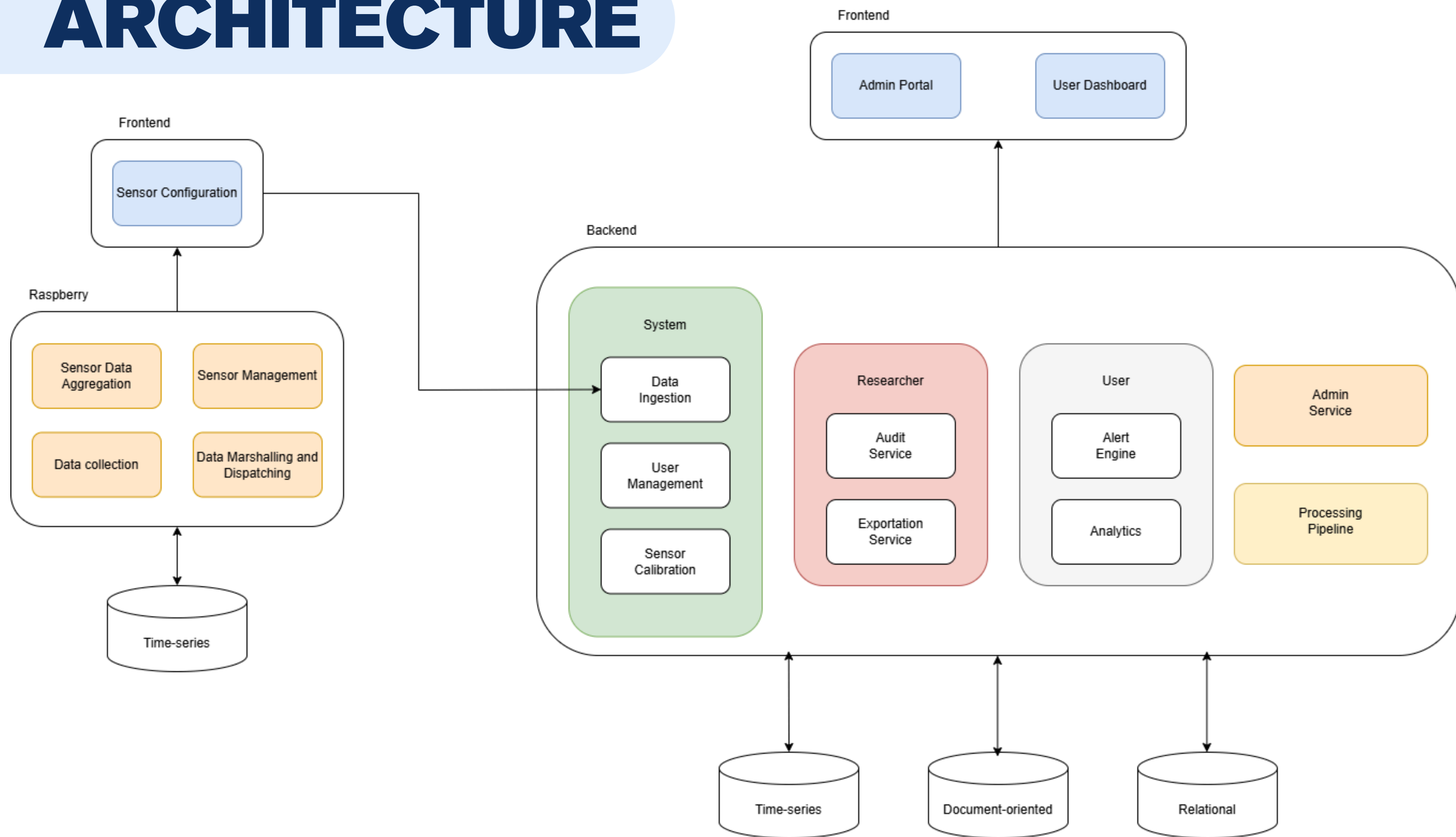
Participants performing repetitive bending movements show 35° average spinal deviation. Fatigue levels increase by 45% after 2 hours.

Movement Break Impact

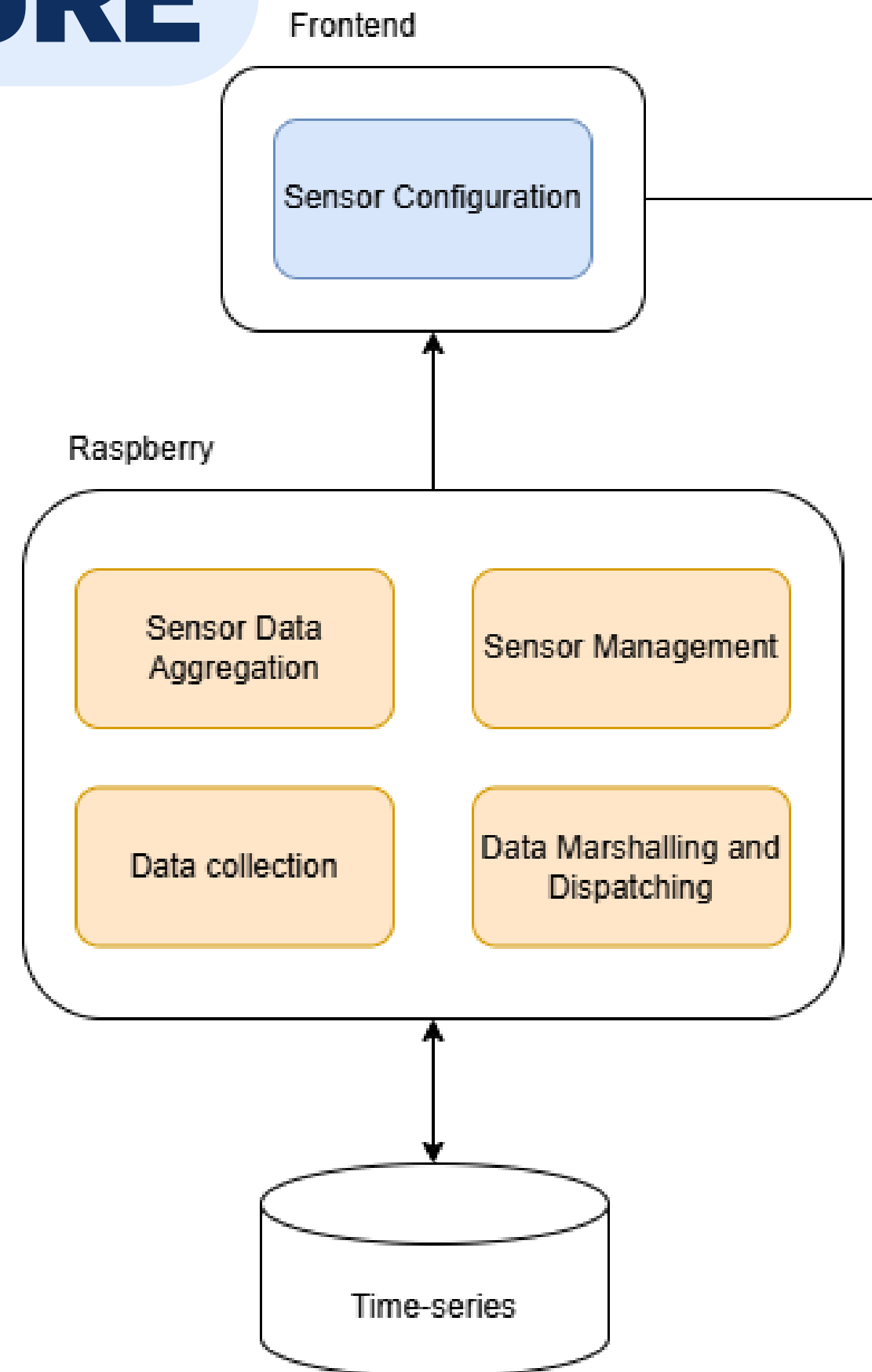
Participants with scheduled breaks every 30 minutes maintain 23% better posture scores compared to control group.

22

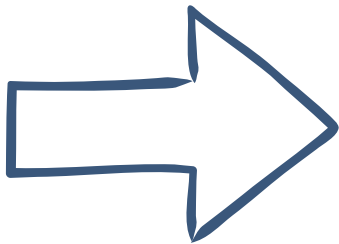
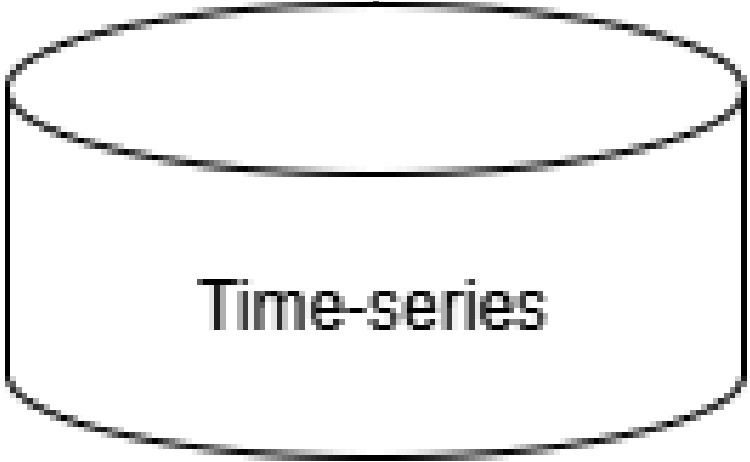
ARCHITECTURE



ARCHITECTURE

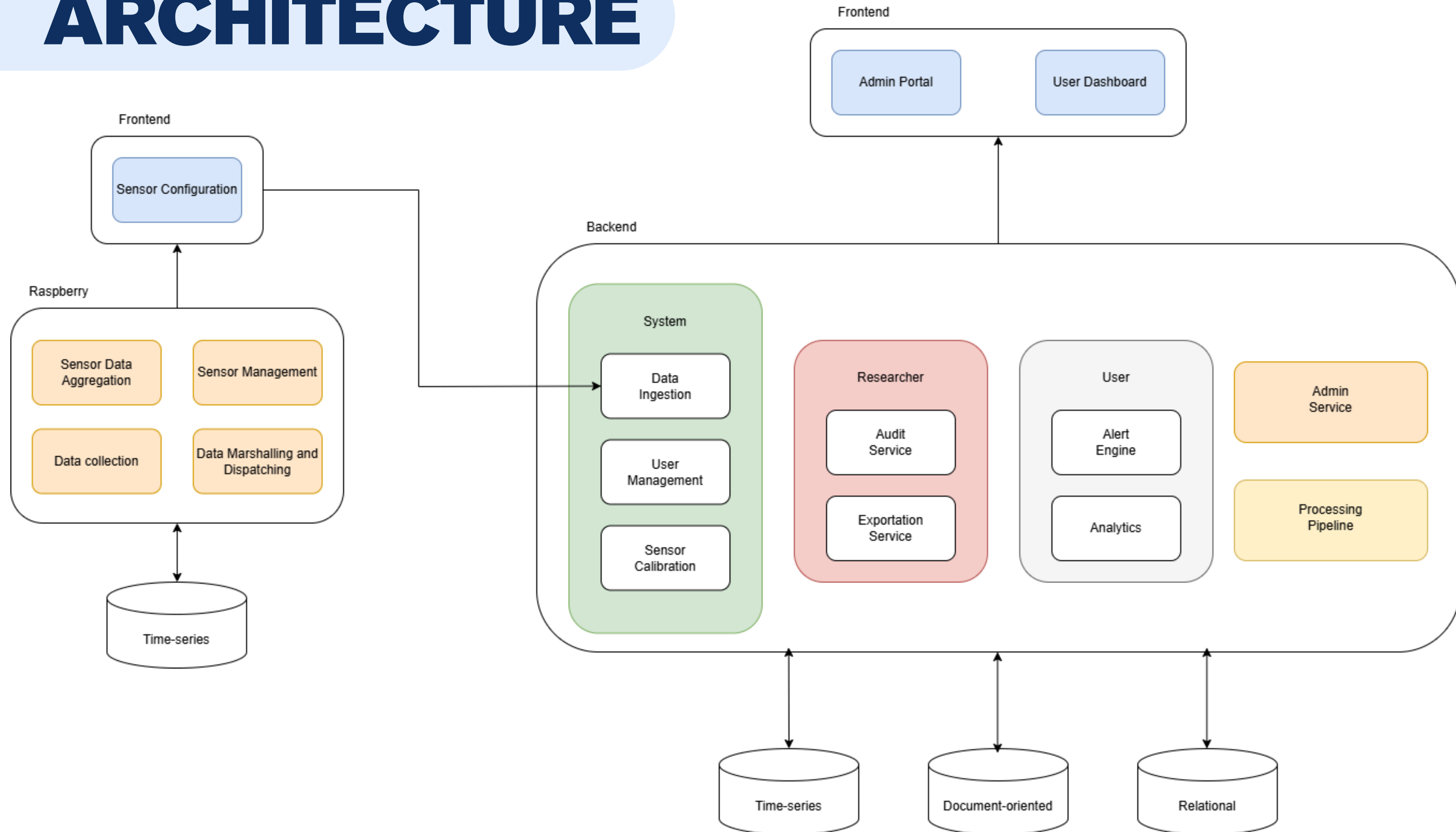


ARCHITECTURE

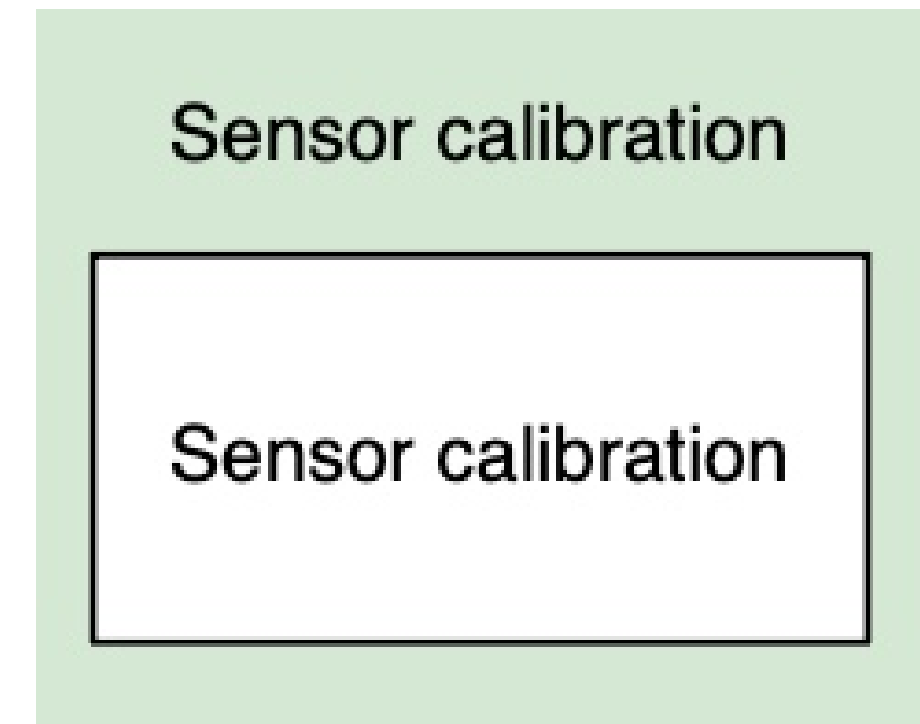
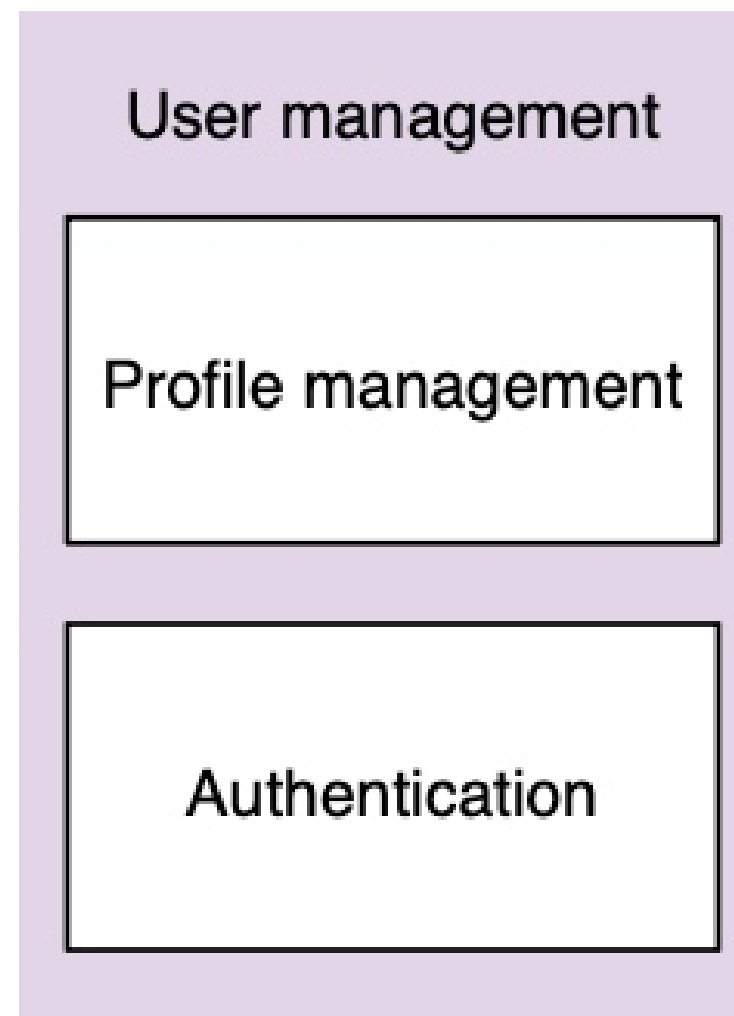
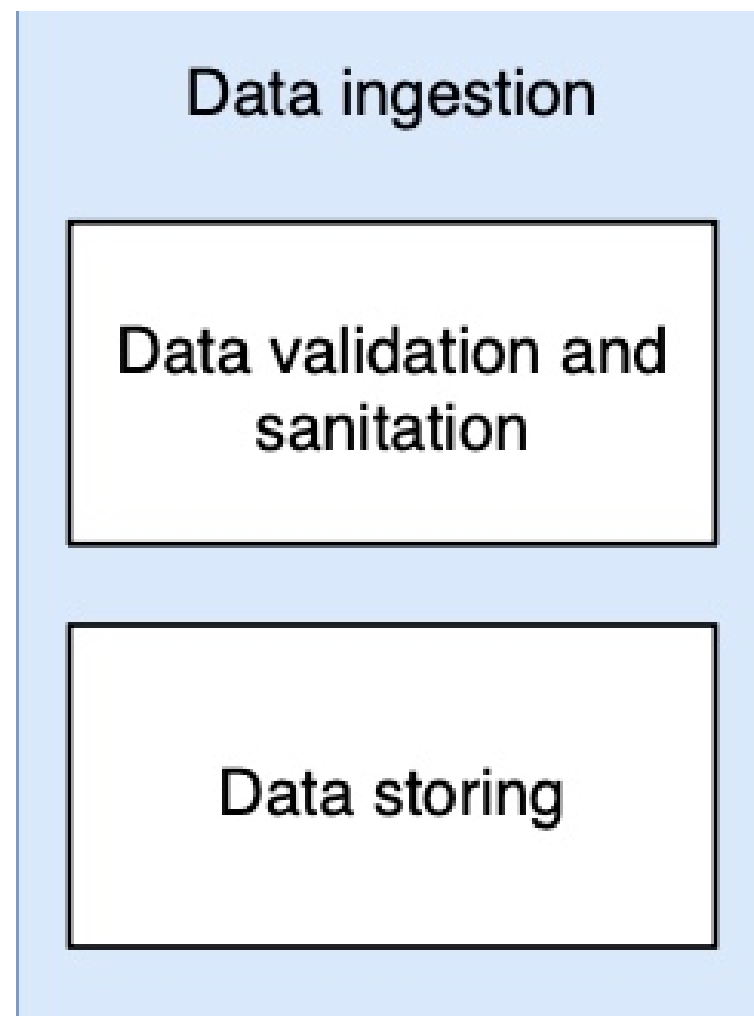


Sensor Data
<u>SensorDataID</u>
SessionID
Timestamp
Quaternion
Acceleration
Gyroscope
CalibrationSys
CalibrationGyro
CalibrationAccel
Euler
LinearAccel
Gravity
Temperature

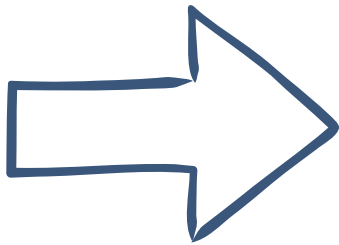
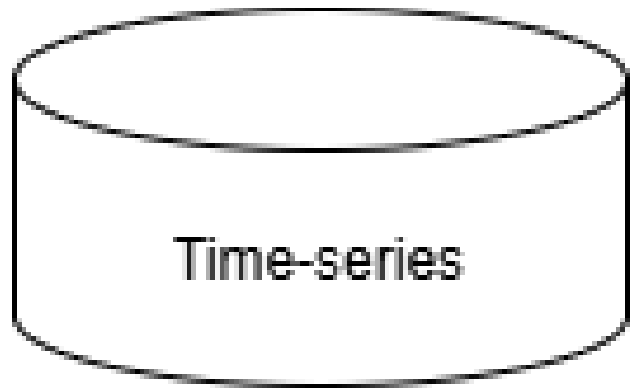
ARCHITECTURE



ARCHITECTURE



ARCHITECTURE

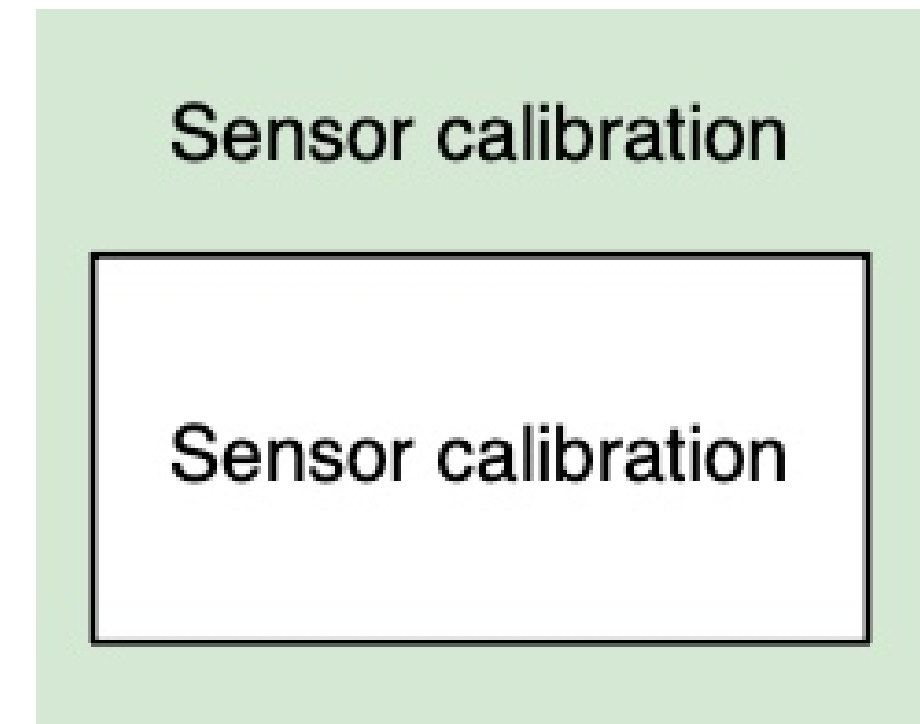
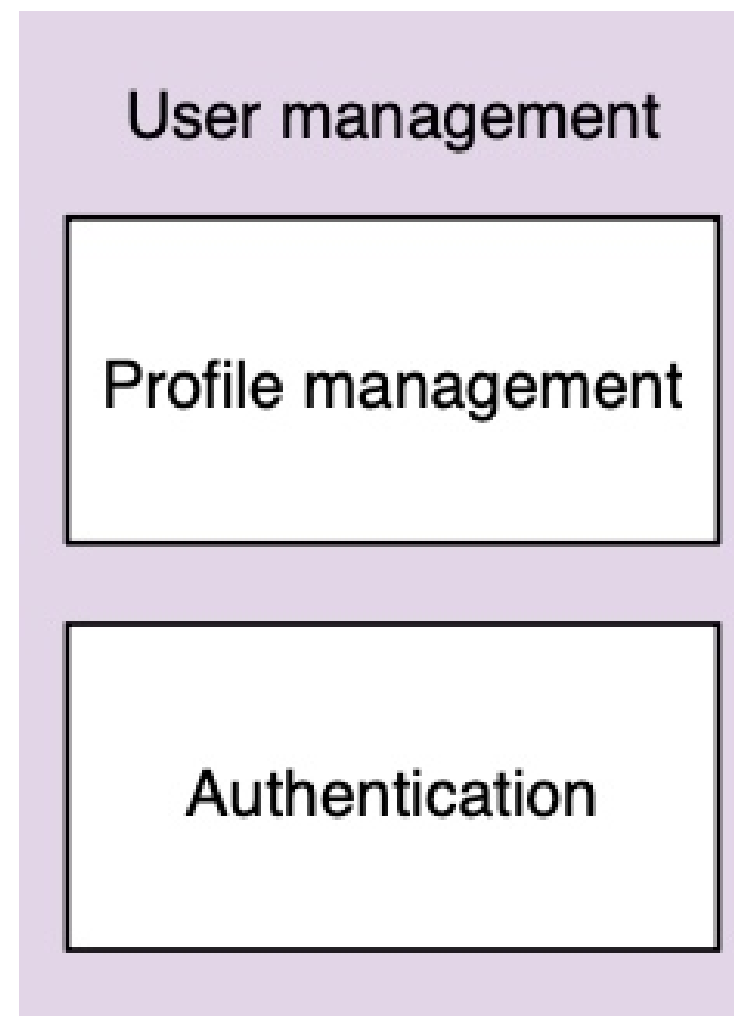
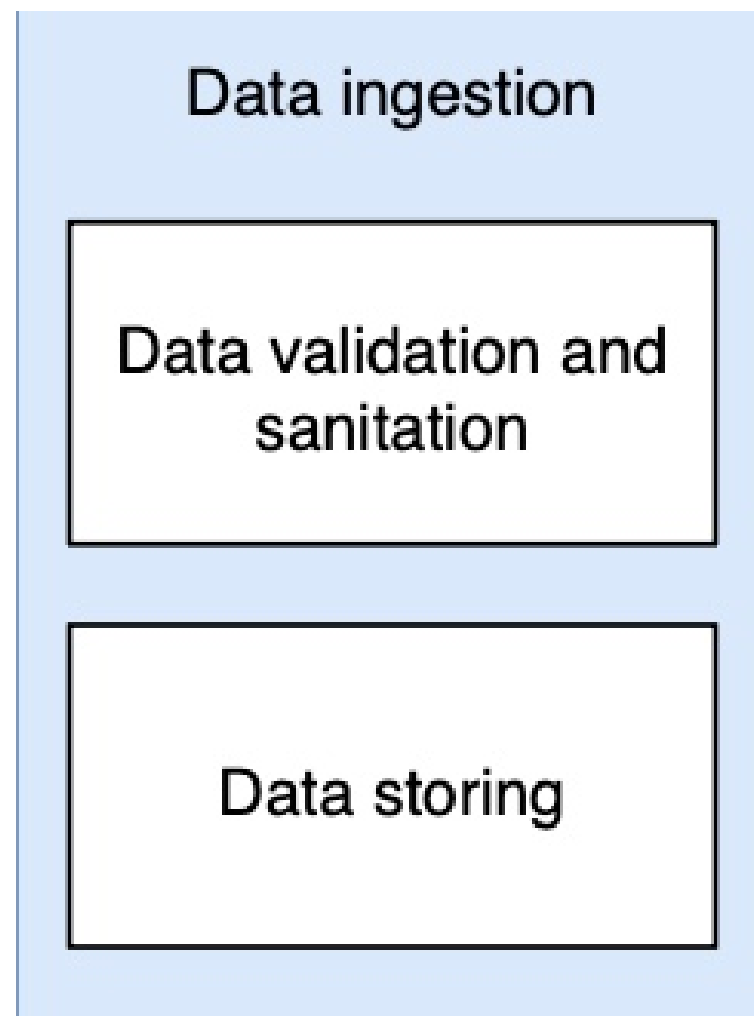


Sensor Data
<u>SensorDataID</u>
SessionID
Timestamp
Quaternion
Acceleration
Gyroscope
CalibrationSys
CalibrationGyro
CalibrationAccel
Euler
LinearAccel
Gravity
Temperature
IsValid

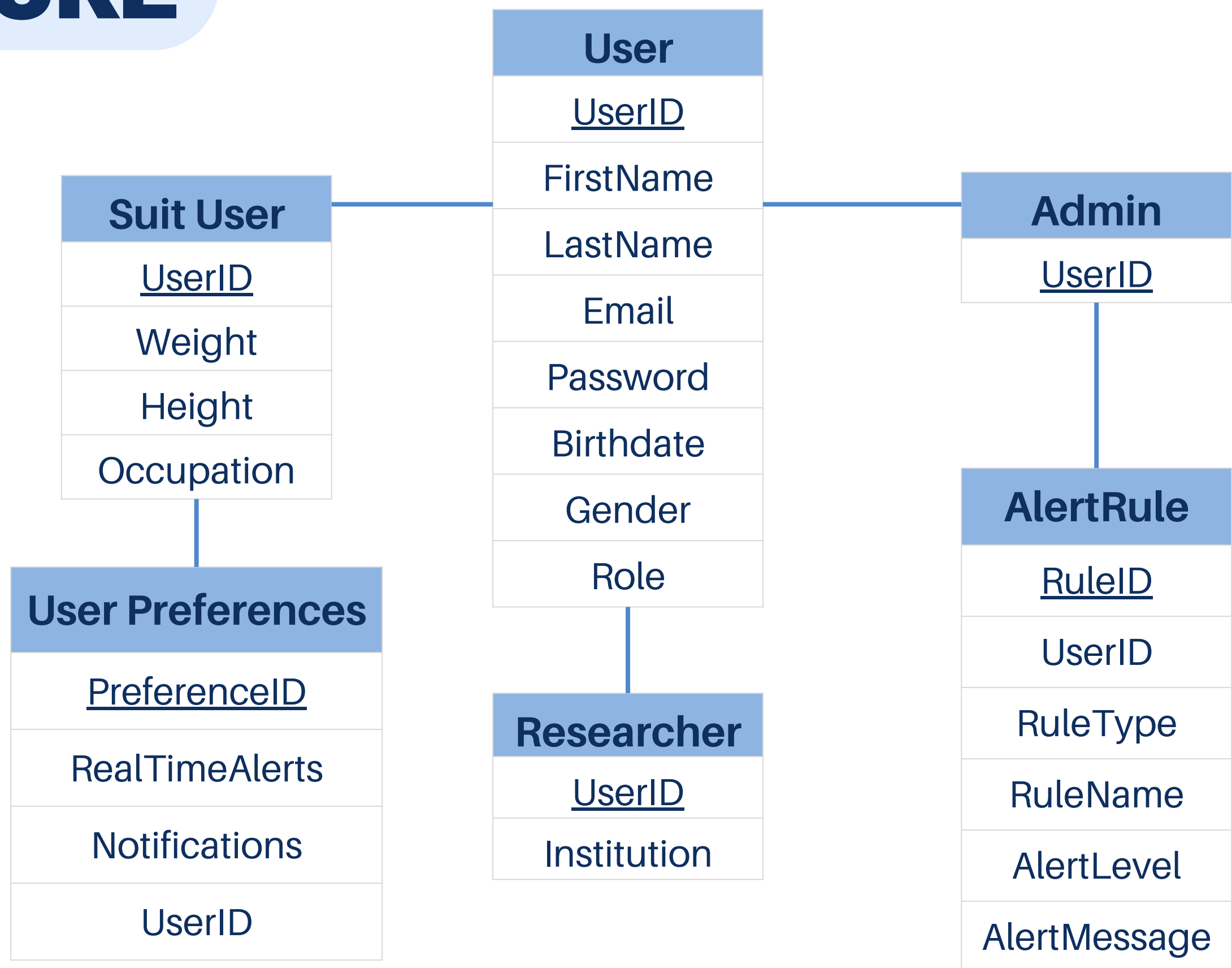
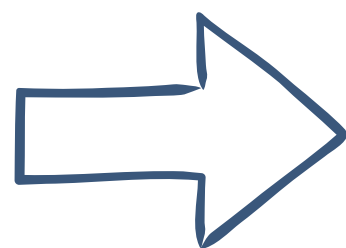
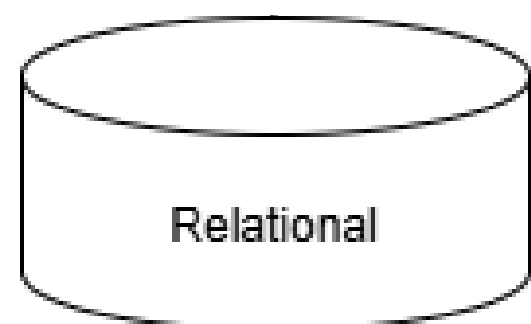
Session
<u>SessionID</u>
UserID
TimestampStart
TimestampEnd
AvgPostureQuality

Posture Alert
<u>SessionID</u>
AlertID
AlertType
Severity
Description
Timestamp

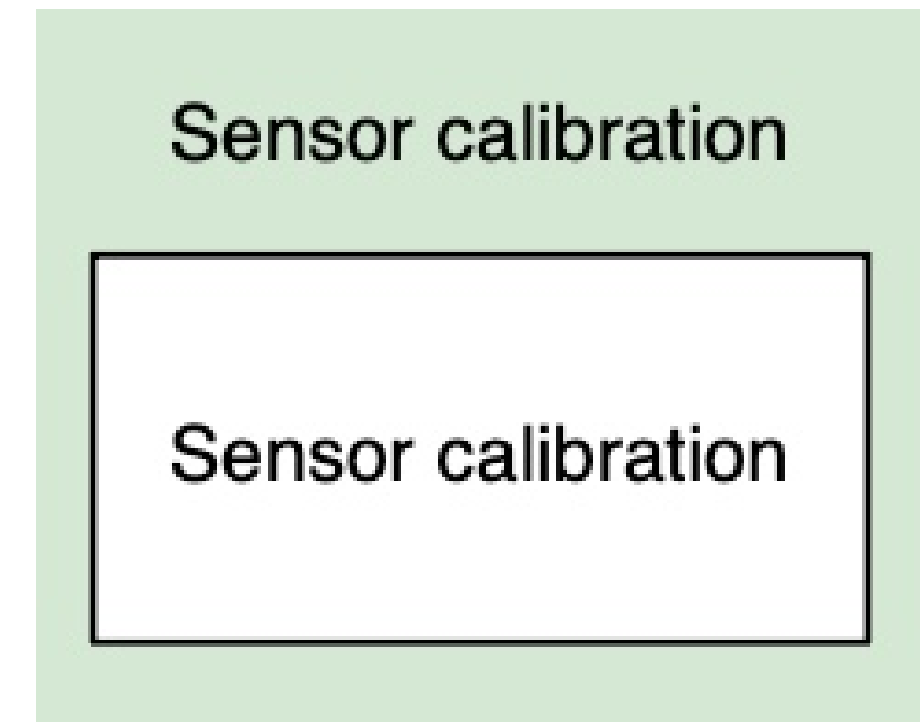
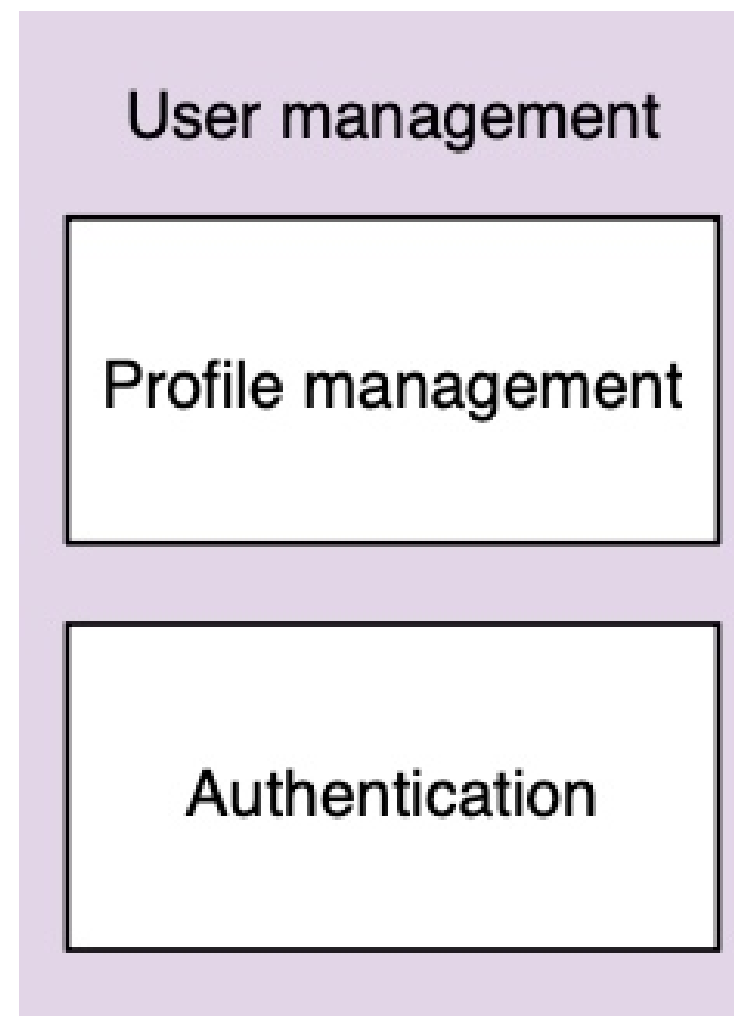
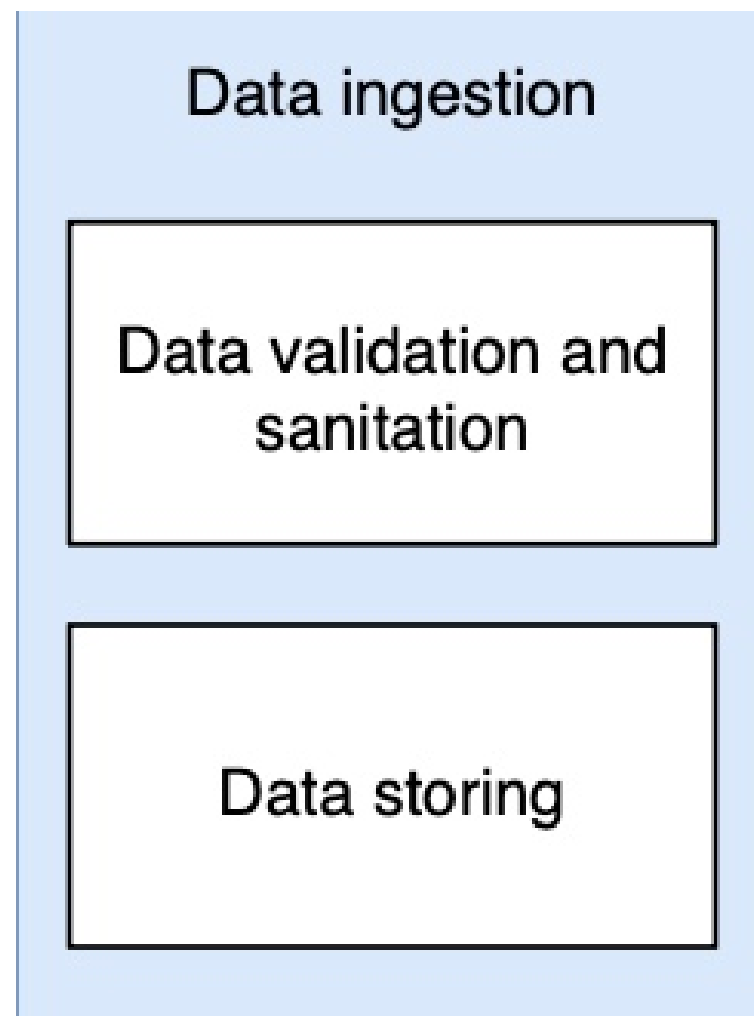
ARCHITECTURE



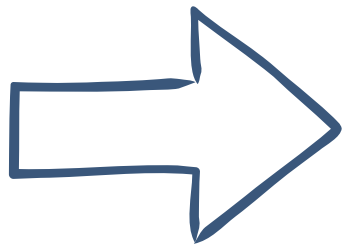
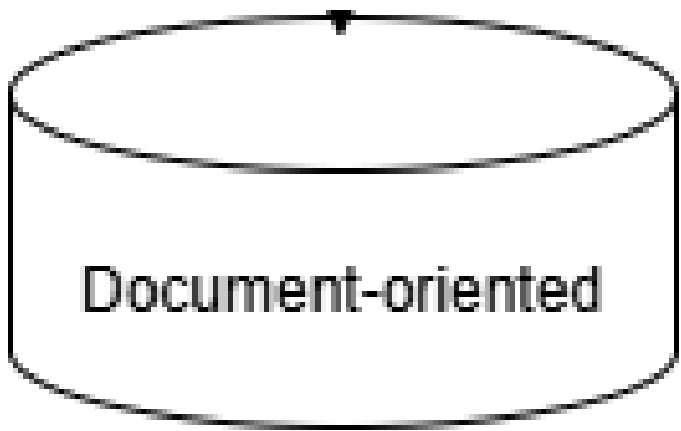
ARCHITECTURE



ARCHITECTURE

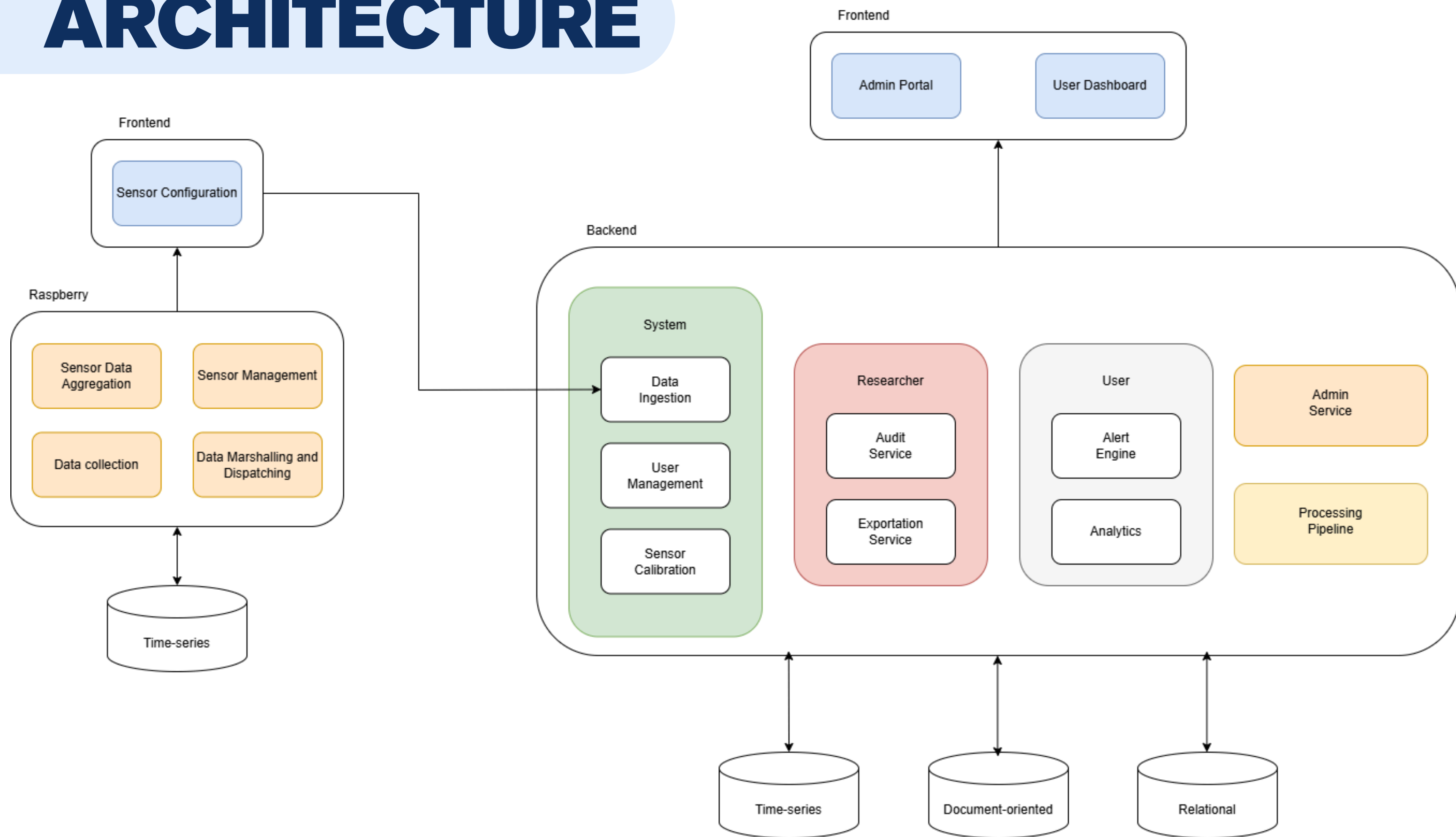


ARCHITECTURE

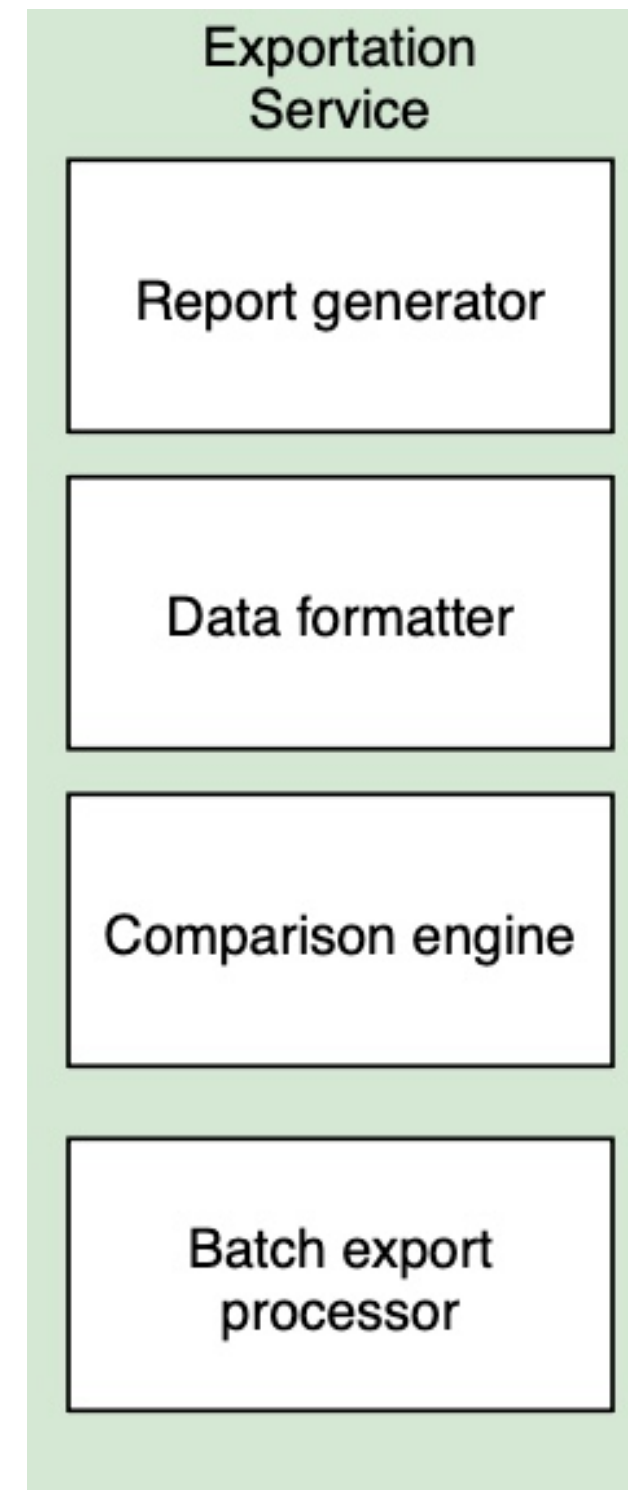


User Baseline
<u>BaselineID</u>
UserID
LastCalculatedAt
FatiguePercentage
IdealPosturePercentage
ShoulderAngleLimit
KneeAngleLimit
BackAngleLimit
NeckAngleLimit

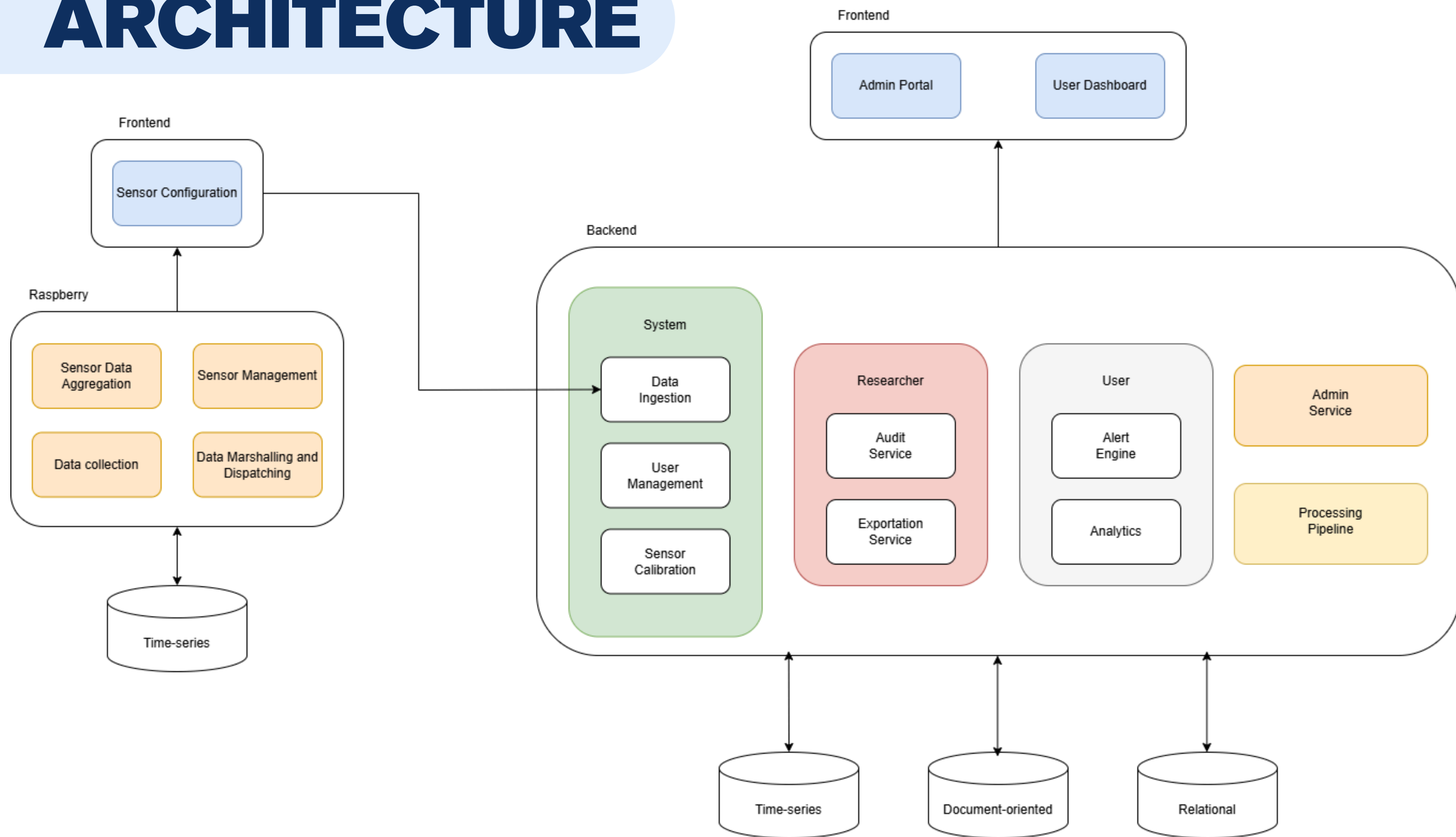
ARCHITECTURE



ARCHITECTURE



ARCHITECTURE



ARCHITECTURE

Alert Engine

Rule Manager

Threshold Calculator

Notification
Dispatcher

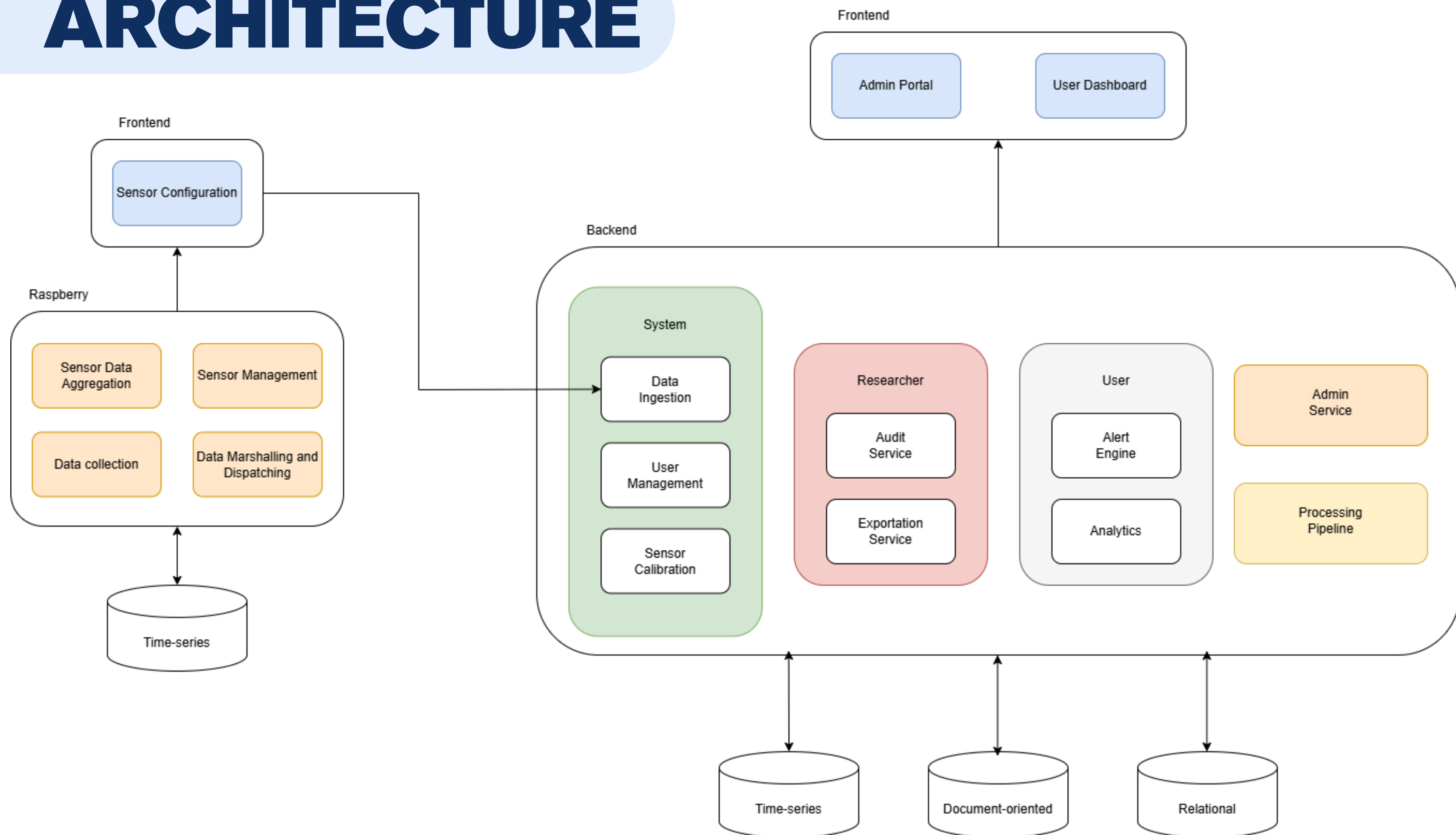
Analytics

Trend analysis

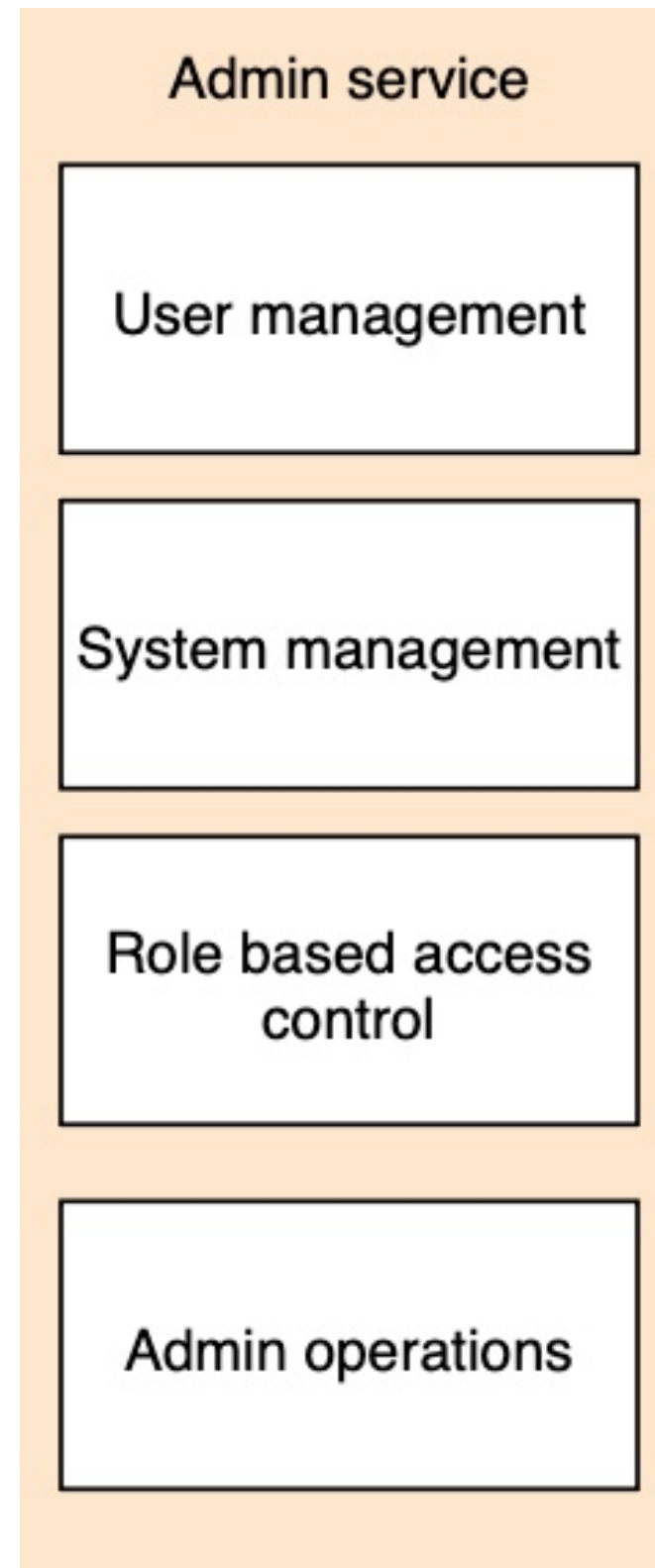
Statistical
aggregation

Report generation

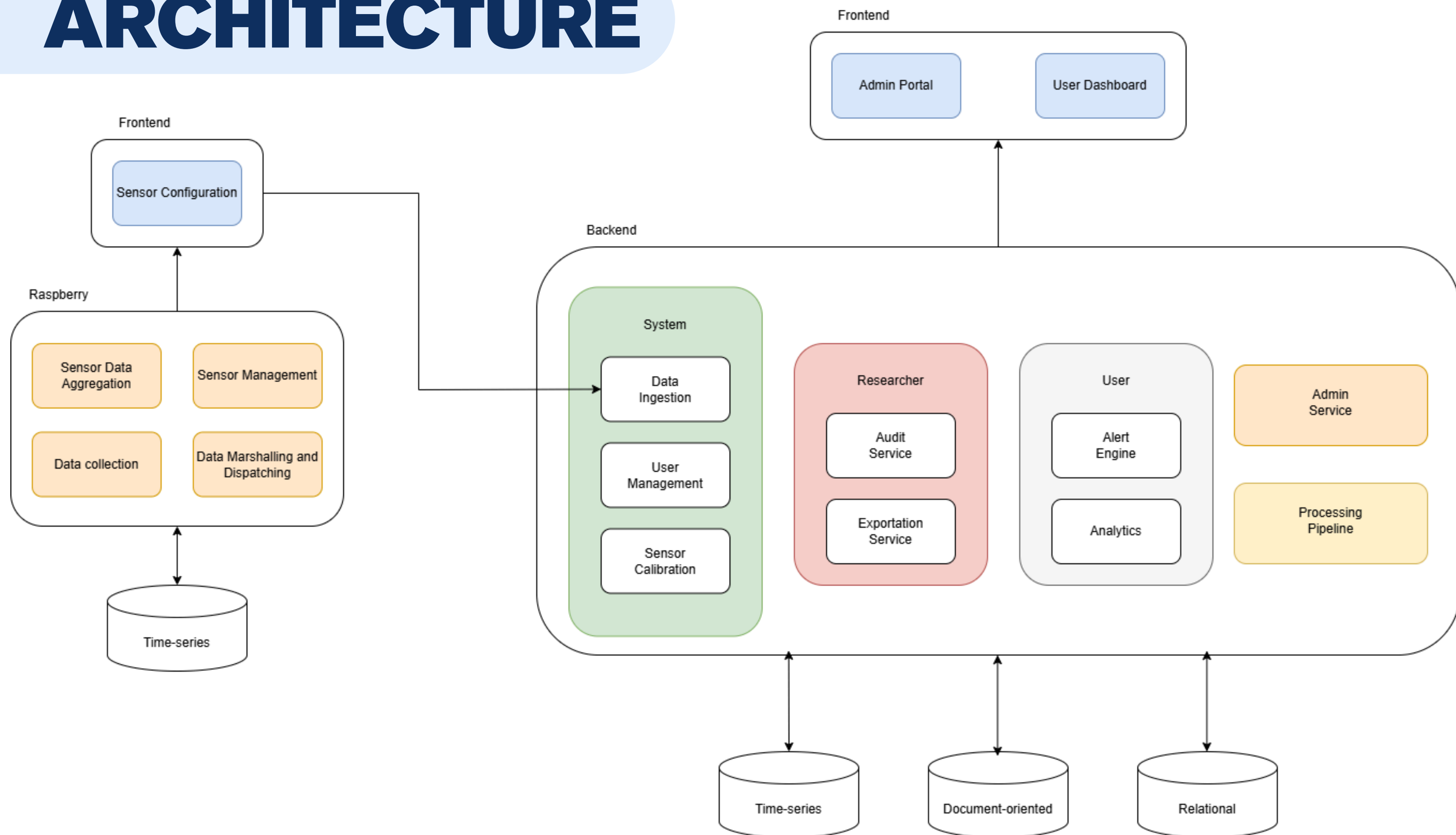
ARCHITECTURE



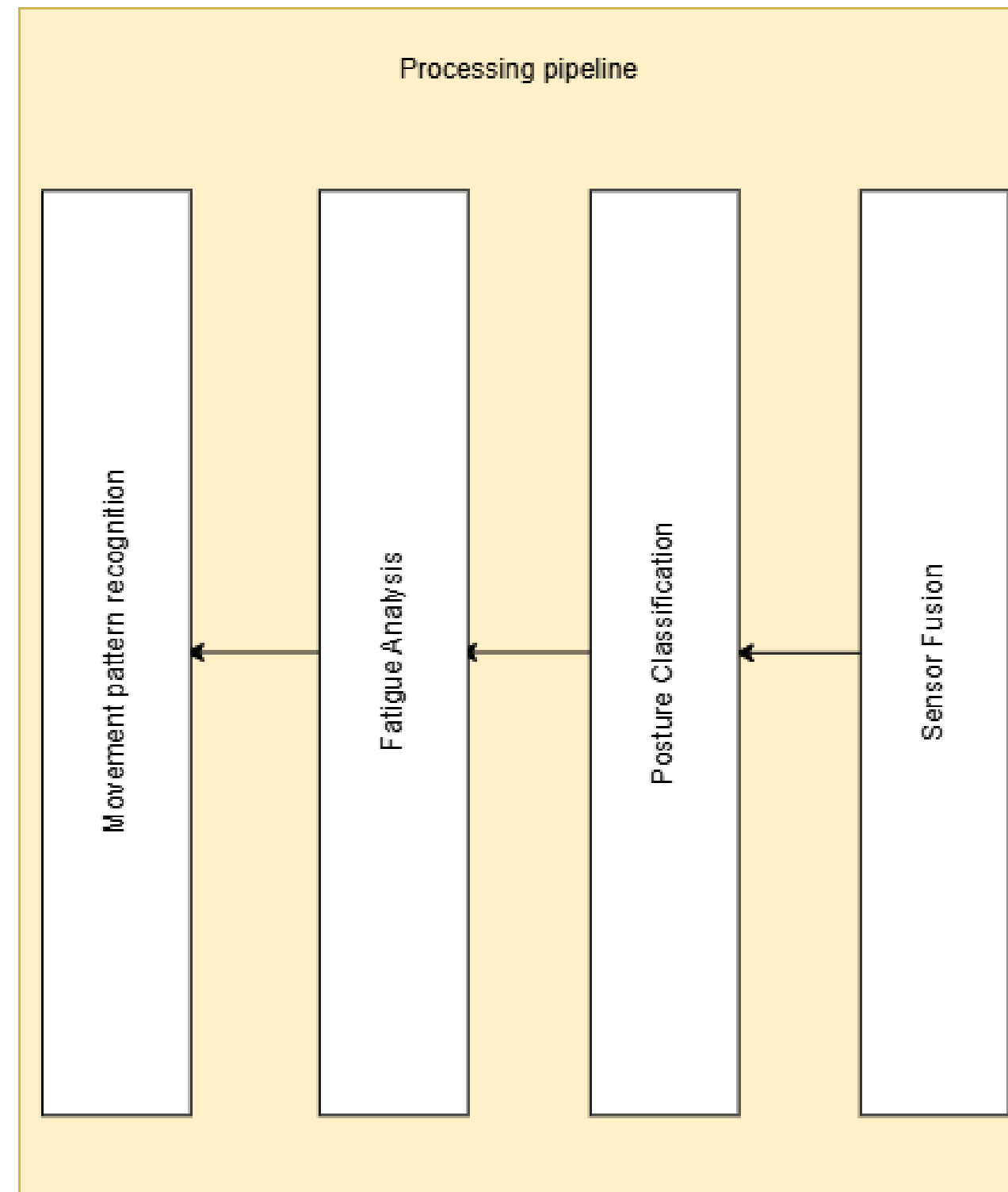
ARCHITECTURE



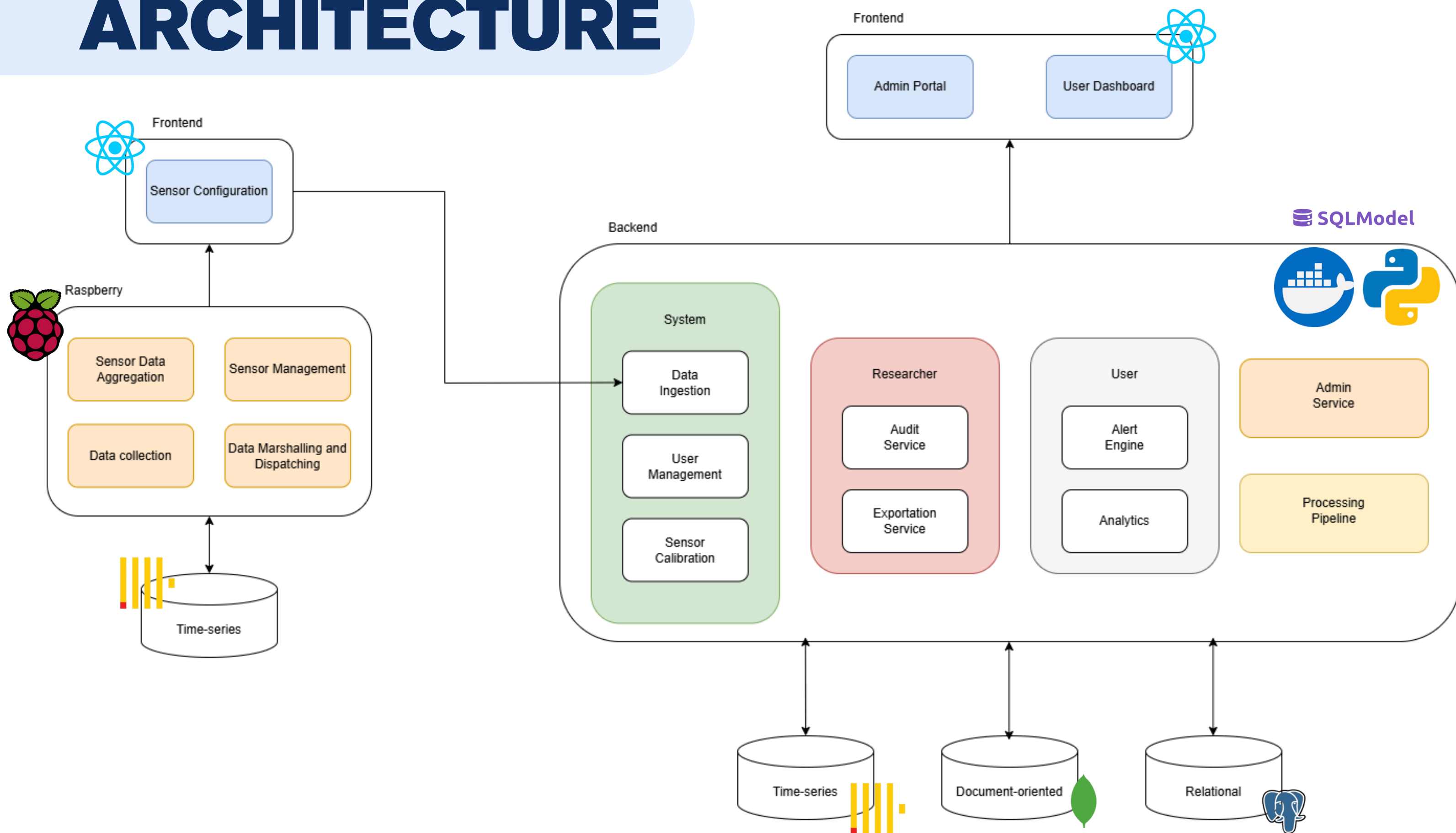
ARCHITECTURE



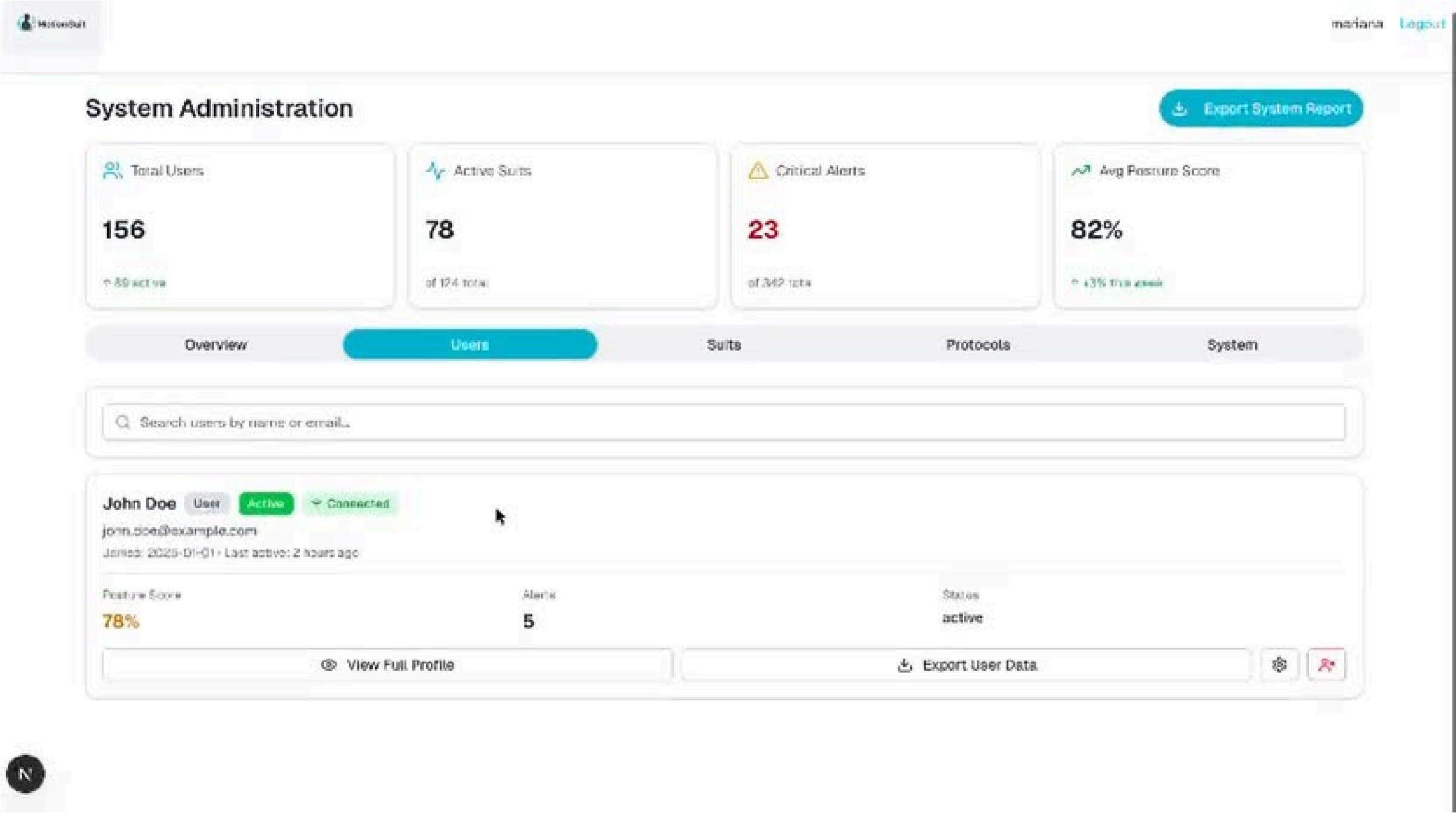
ARCHITECTURE



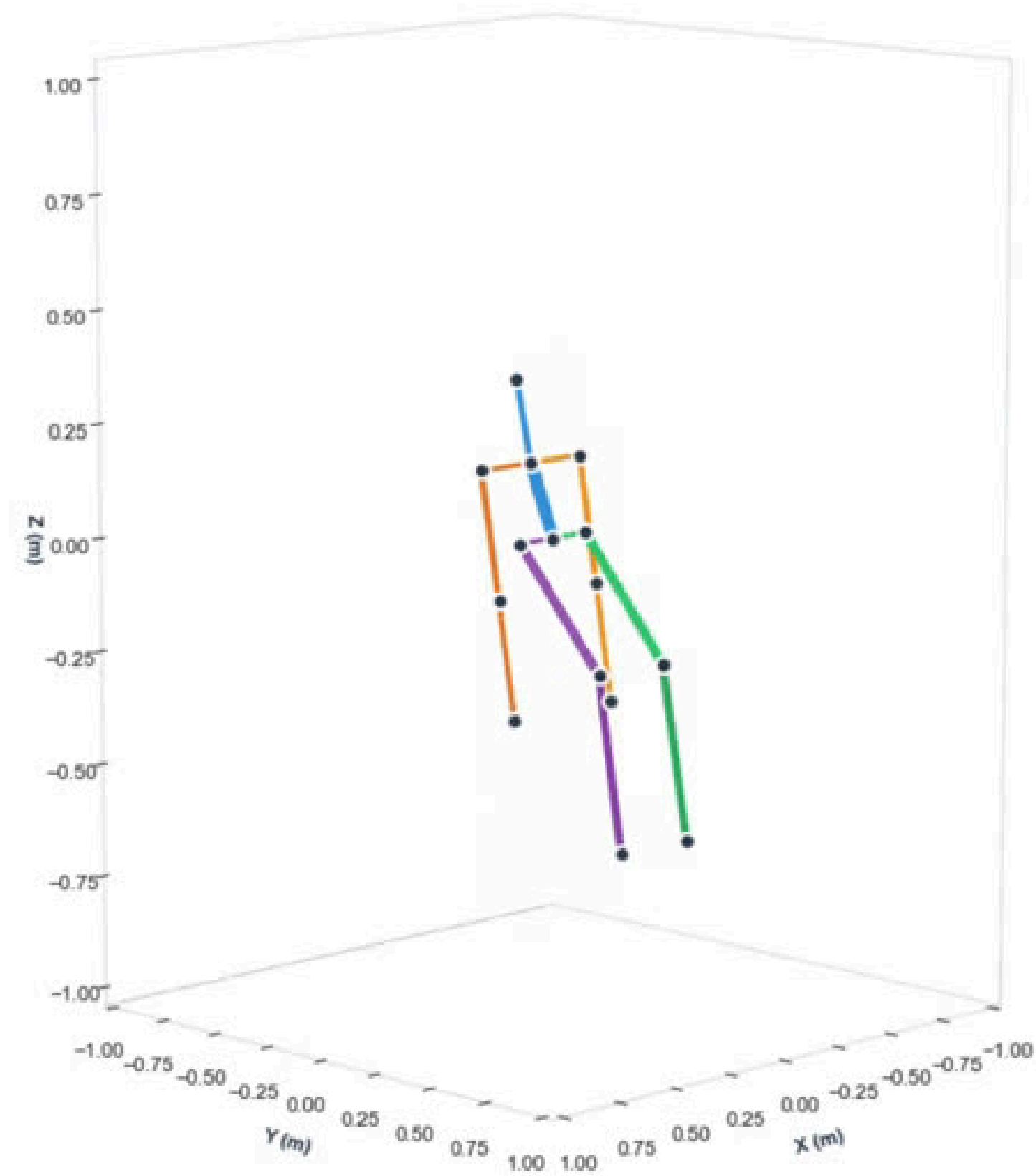
ARCHITECTURE

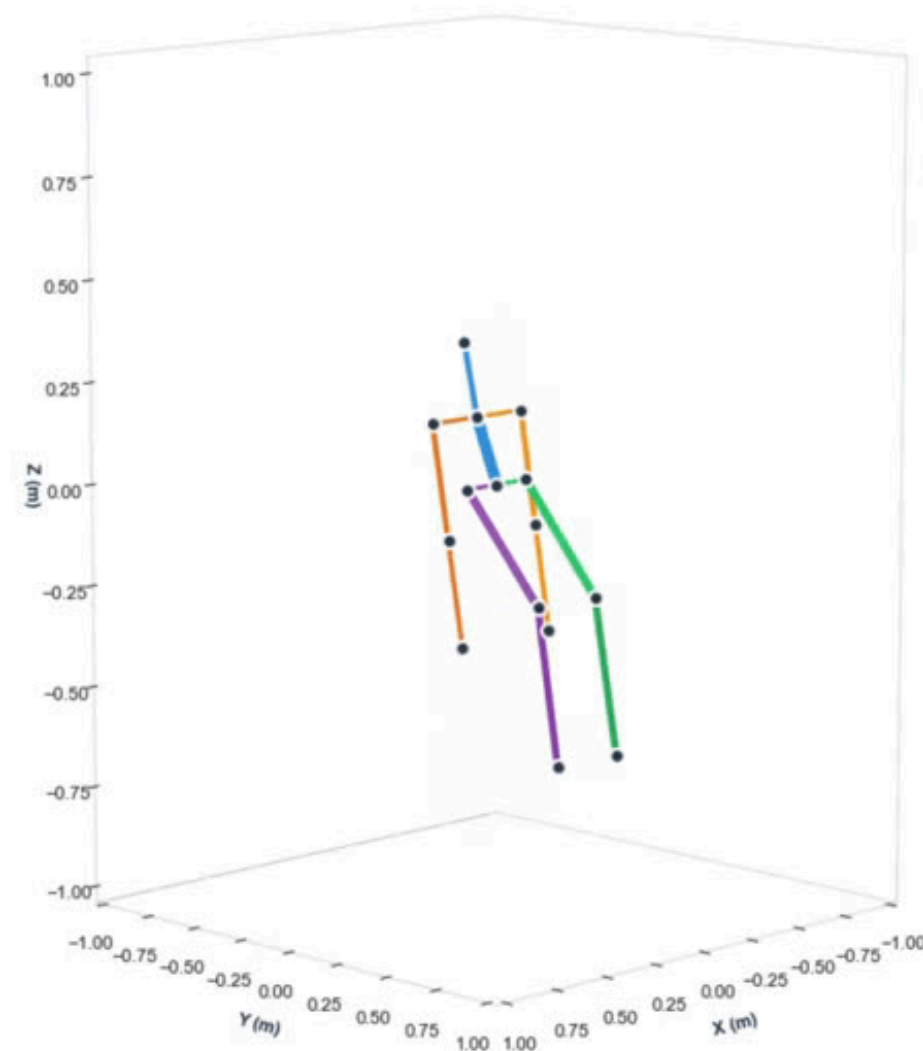


FRONTEND



SENSORS





THANK YOU!



<https://pei-motionsuit.github.io/microsite/>